

深度学习环境配置教程

《参考使用手册》
深度学习实践班

创新创业学院

2021 年 11 月

目 录

一、Windows 系统下深度学习环境配置	2
1.1 Anaconda 的安装	2
1.1.1 Anaconda 简介	2
1.1.2 Anaconda 安装演示	3
1.2 Conda 创建虚拟环境	13
1.3 PyCharm 的安装和环境、解释器配置	16
1.4 CUDA 的安装	27
1.5 cuDNN 神经网络库的配置	38
1.6 PyTorch 的安装	41
1.7 TensorFlow 的安装	49
二、Windows 系统下深度学习环境配置答疑	49
三、Ubuntu 系统下深度学习环境配置	49

一、Windows 系统下深度学习环境配置

本教程包含深度学习环境配置 Anaconda、PyCharm、CUDA、cuDNN、PyTorch、Tensorflow 的安装和配置，此文档整理主要参考以下视频，同学们可以同步观看视频进行学习。

参考视频网址如下：

[https://www.bilibili.com/video/BV1ov41137Z8?from=search&seid=15663295424584842751
&spm_id from=333.337.0.0](https://www.bilibili.com/video/BV1ov41137Z8?from=search&seid=15663295424584842751&spm_id_from=333.337.0.0)

1.1 Anaconda 的安装

1.1.1 Anaconda 简介

Anaconda 官方网址：<https://www.anaconda.com/>

Anaconda 可以便捷获取包且对包能够进行管理，同时对环境可以统一管理的发行版本。

Anaconda 包含了 conda、Python 在内的超过 180 个科学包及其依赖项。

Anaconda 具有如下特点：

- 开源
- 安装过程简单
- 高性能使用 Python 和 R 语言
- 免费的社区支持

其特点的实现主要基于 Anaconda 拥有的：

- conda 包
- 环境管理器
- 1,000+开源库

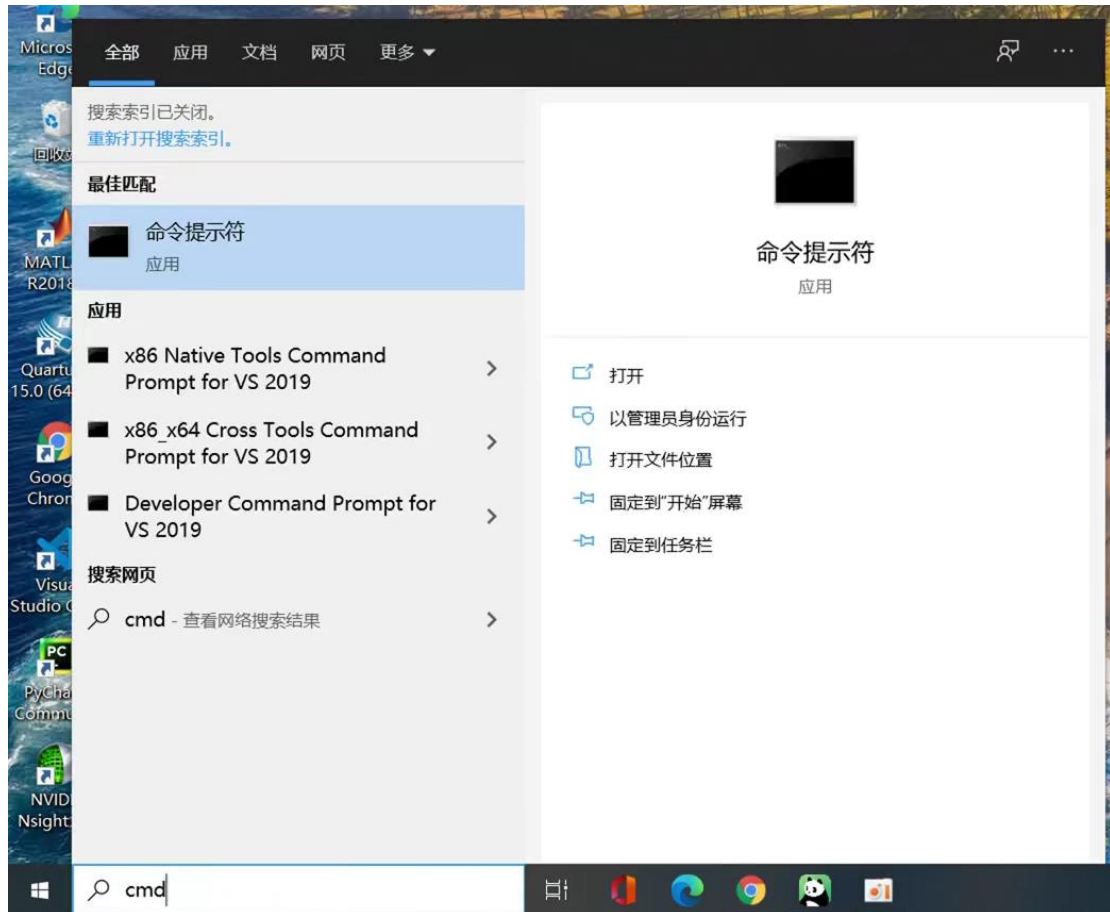
如果日常工作或学习并不必要使用 1,000 多个库，那么可以考虑安装 Miniconda（图形界面下载及命令行安装见：<https://docs.conda.io/en/latest/miniconda.html>），这里不过多介绍 Miniconda 的安装及使用。

【注，以上 Anaconda 简介转载自：<https://www.jianshu.com/p/62f155eb6ac5>】

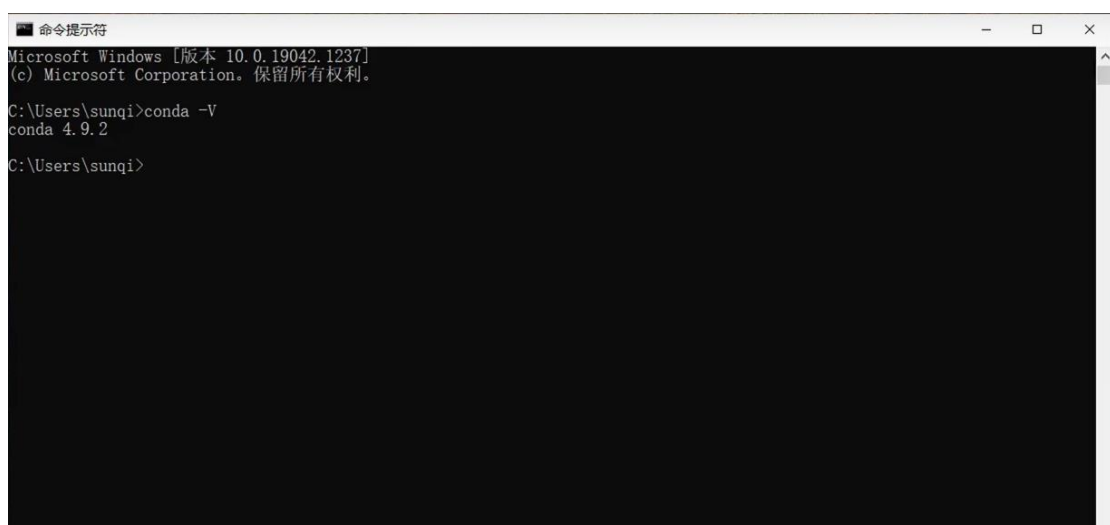
1.1.2 Anaconda 安装演示

1) 打开 cmd，查看是否安装 Anaconda

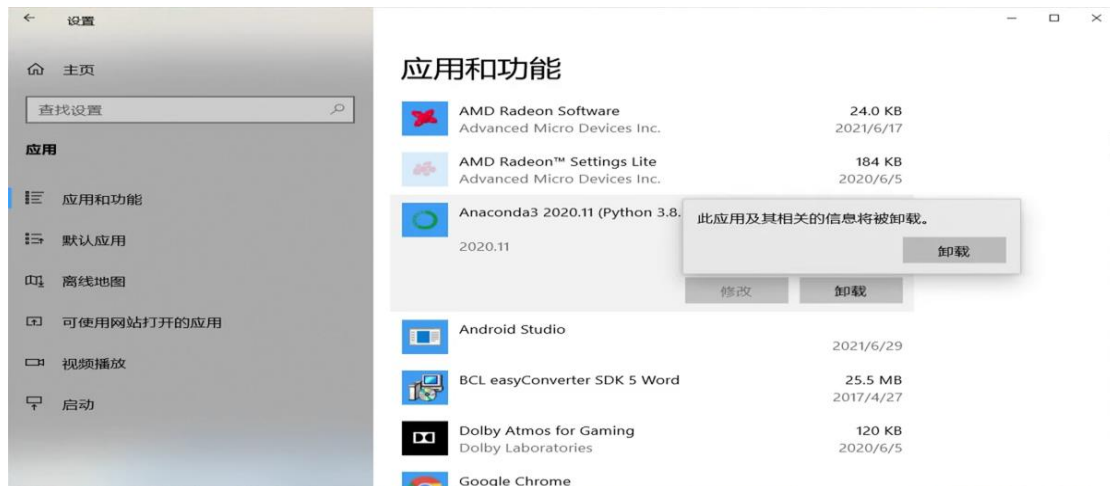
方法：win10 可以通过 win 键+R 打开运行或者开始-运行-输入 cmd 即可打开。



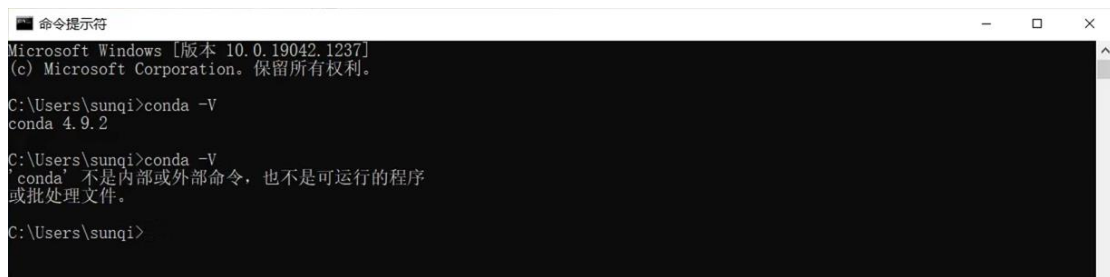
对于已安装过 Anaconda 的，查看 Anaconda 版本的命令是：conda -V



注：若已安装，根据需要可以选择是否卸载后重装新版本，卸载方式：

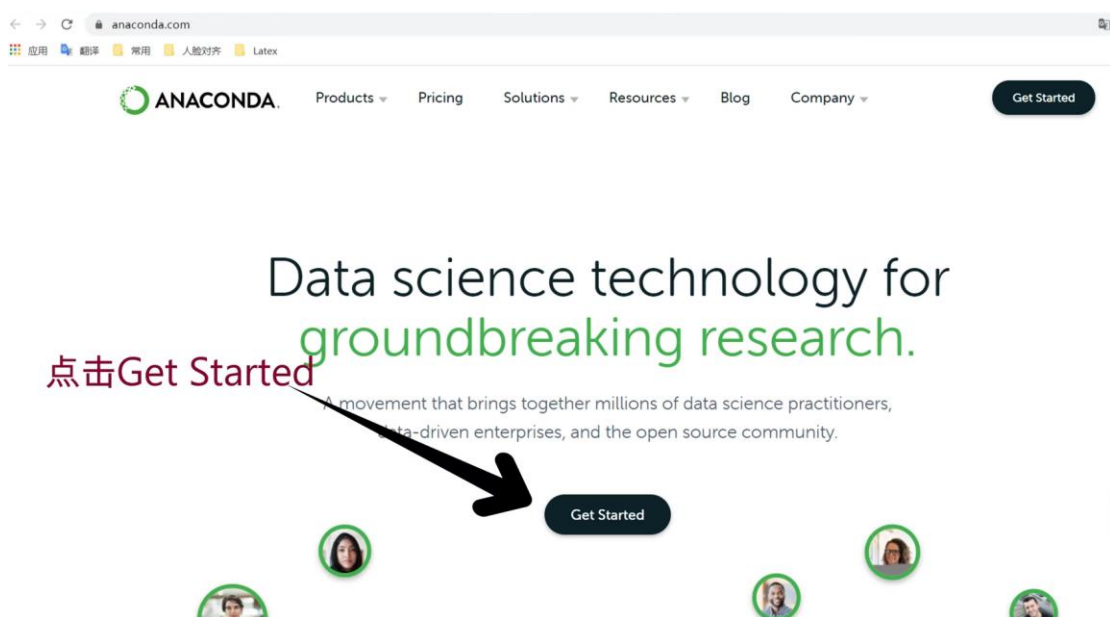


若卸载了 Anaconda 或者未安装 Anaconda，输入命令显示如下：

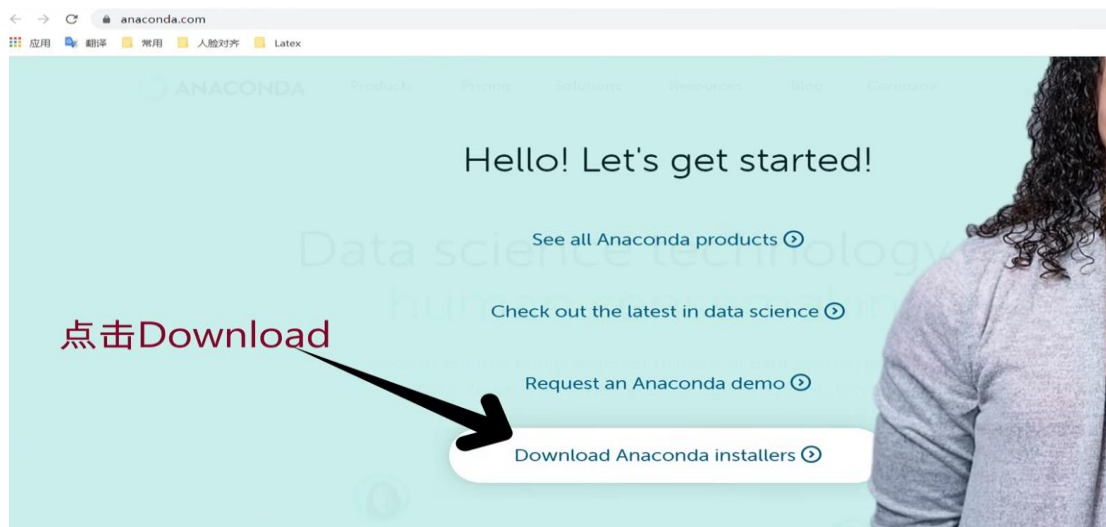


2) 下载 Anaconda

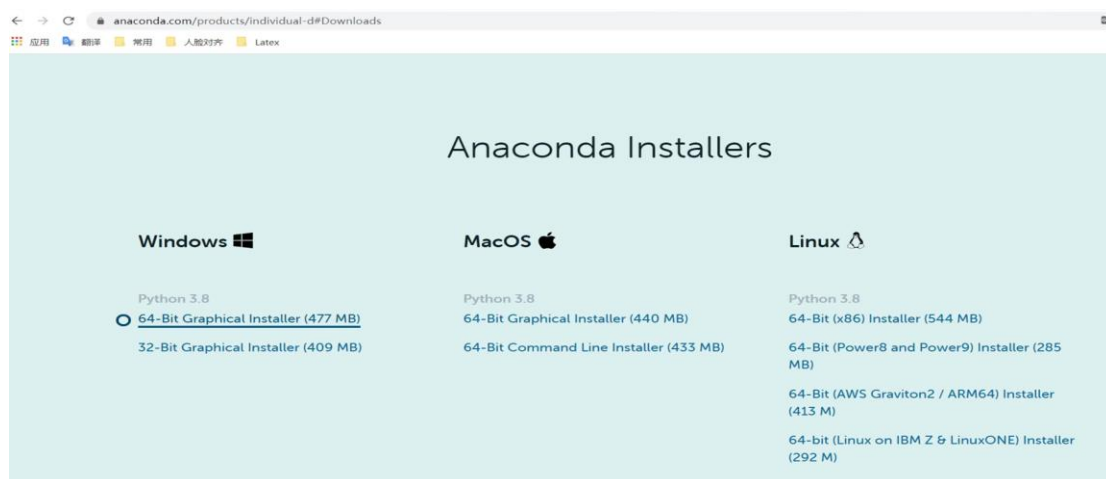
打开 Anaconda 官网 <https://www.anaconda.com/>，点击“Get Started”。



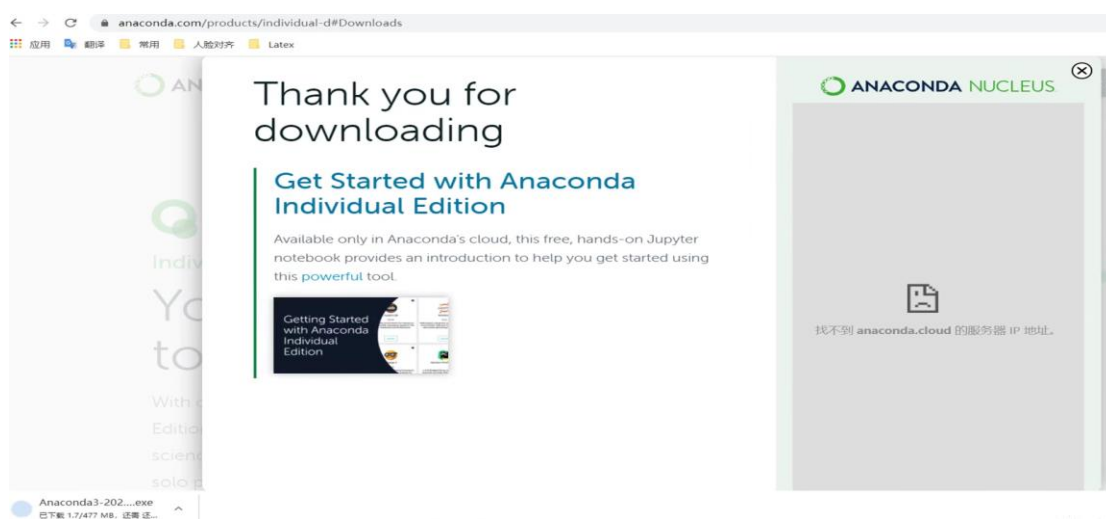
点击“Download Anaconda installers”。



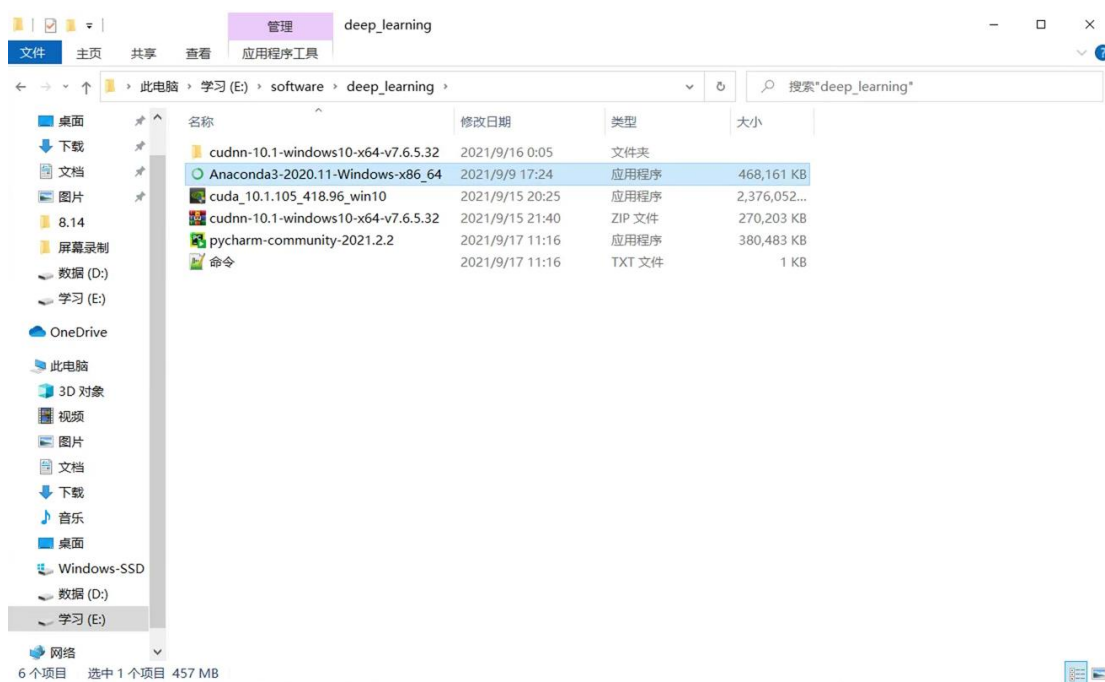
选择所需的版本下载，如



下载中……

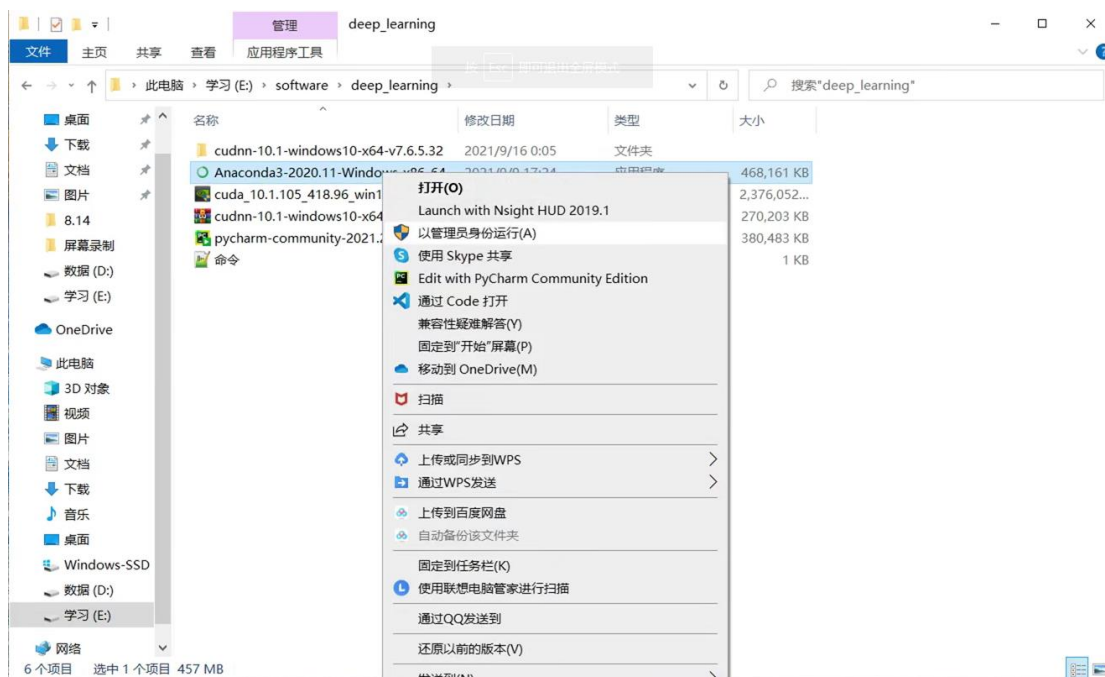


下载后，存放在本地硬盘中，如

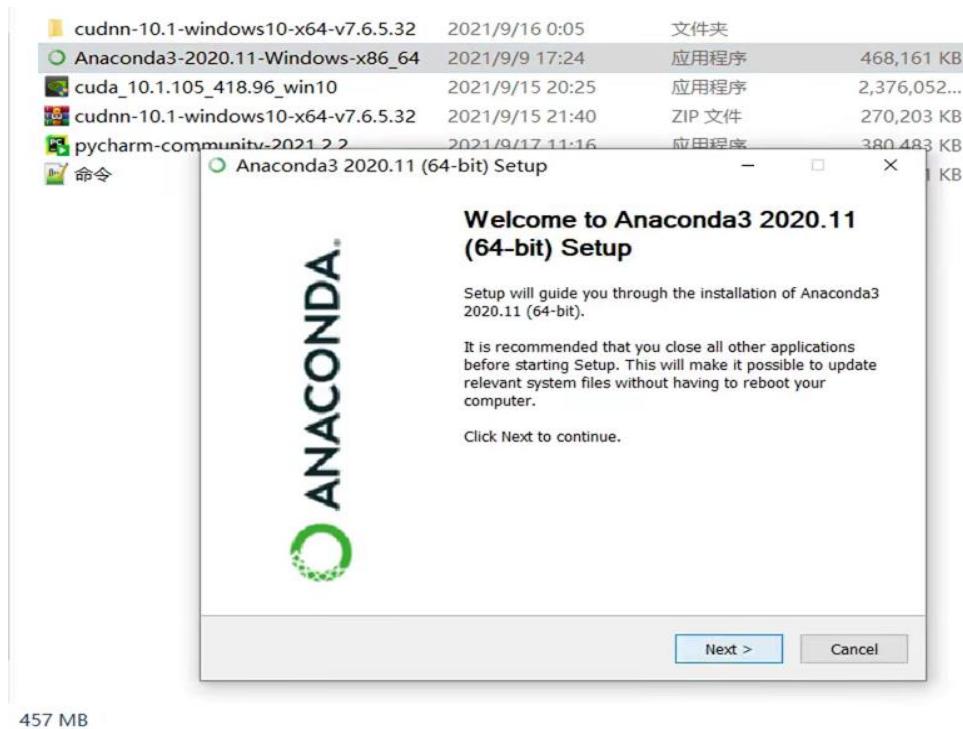


3) 安装 Anaconda

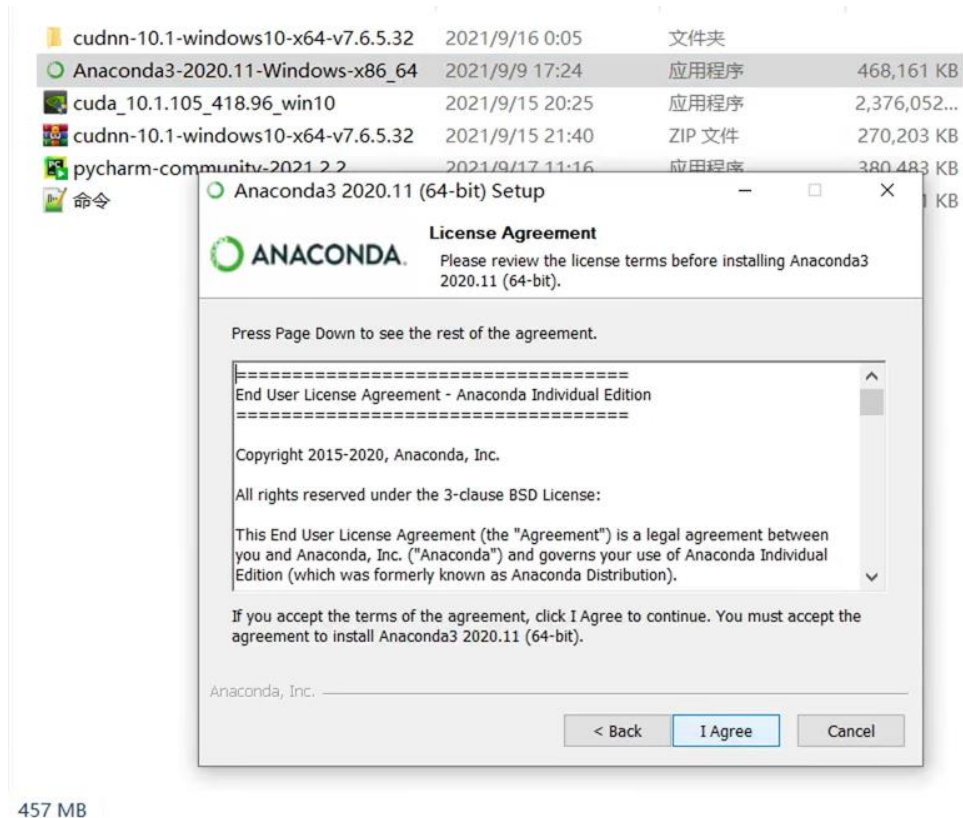
右击 “Anaconda3-2020.11-Windows-x86_64” 应用程序，选择 “以管理员身份运行”，



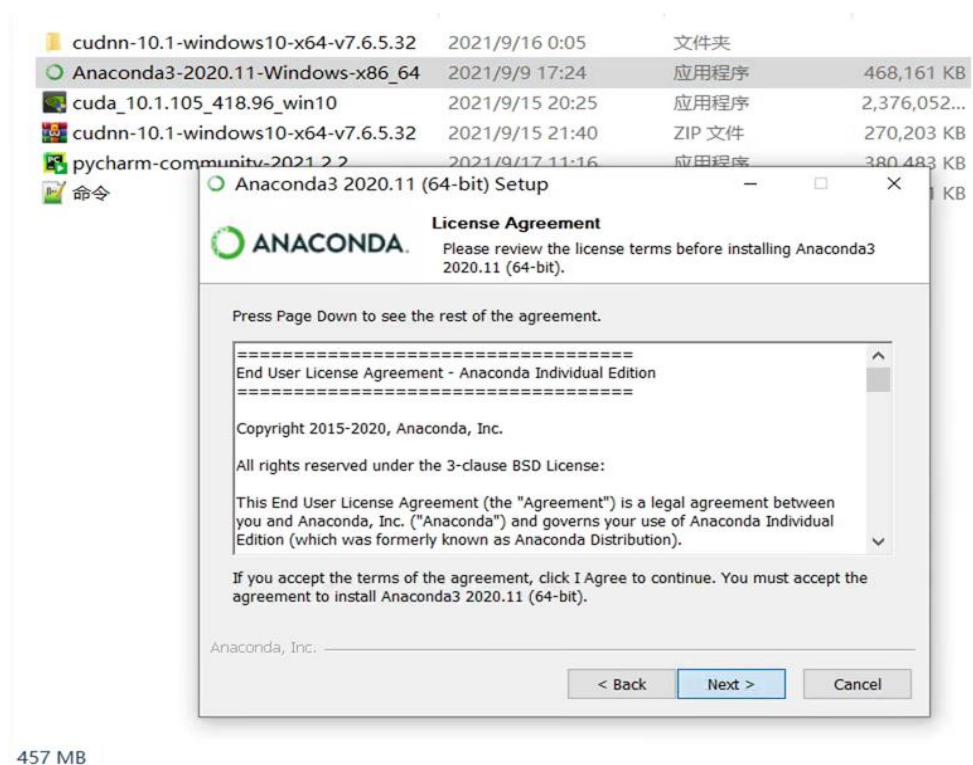
点击“next”，



点击“I Agree”，

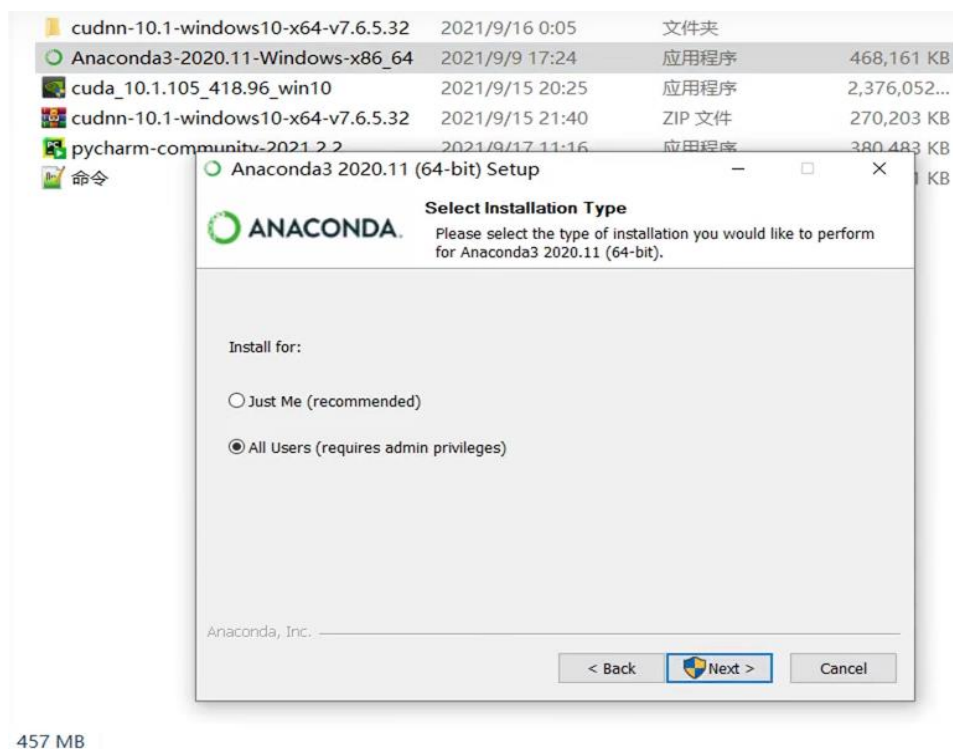


点击“Next”，

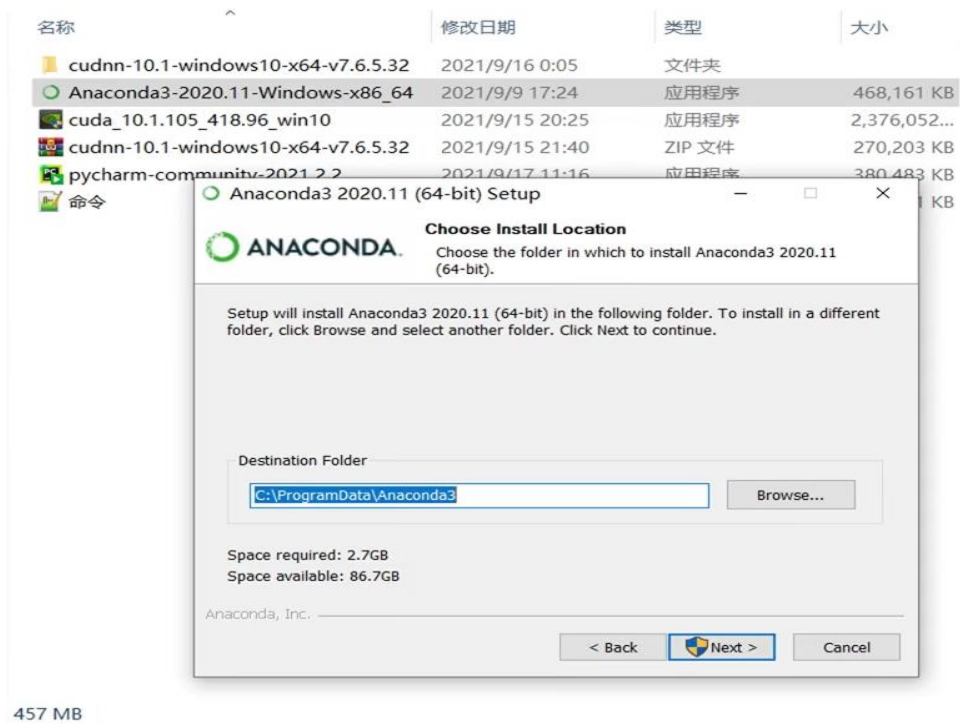


点击“Next”，

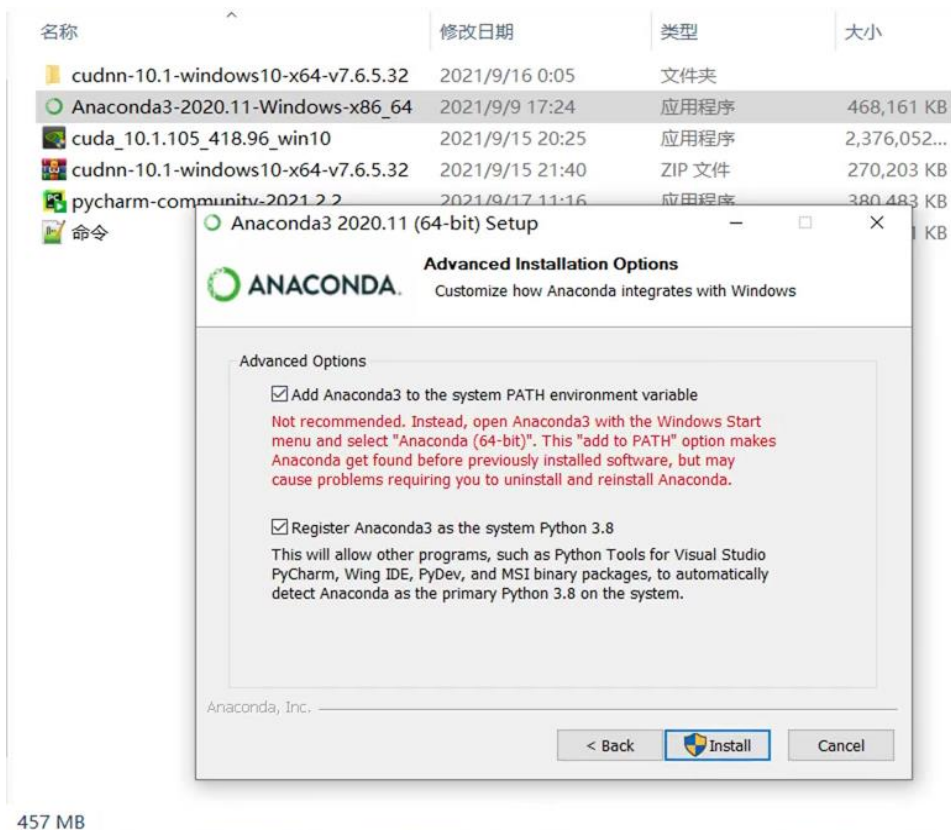
【注：此处选择“Just Me”还是“ALL Users”根据所配机器是否为仅供个人用户使用而定，不影响配置环境。此处以个人机器为例，选择“All Users”。】



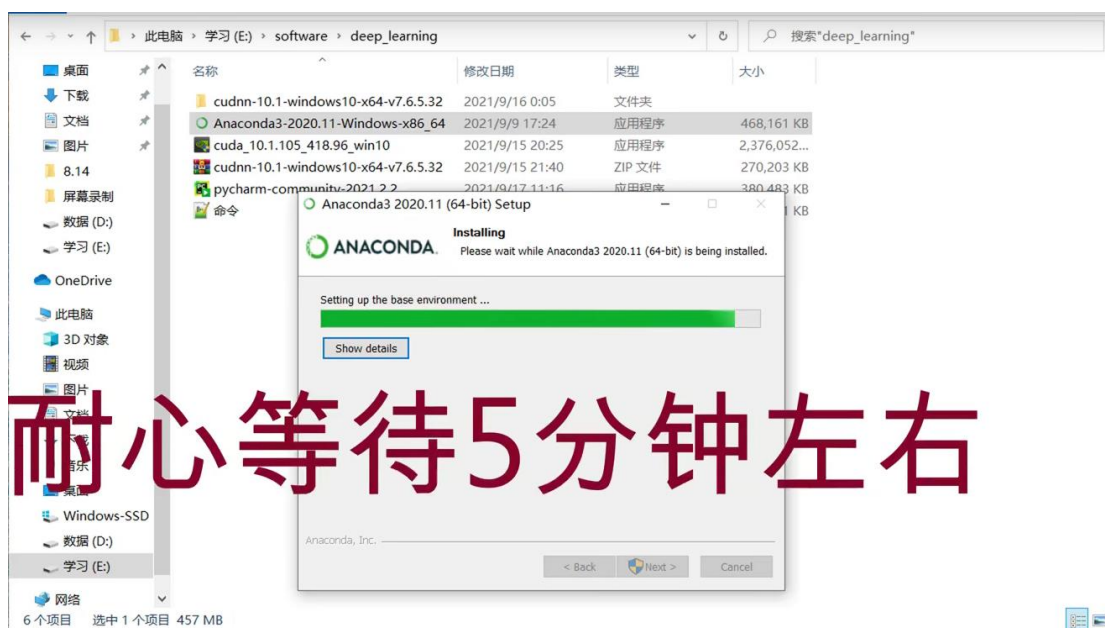
选择安装路径，默认 C 盘或者改为 D 盘也可以，点击 “Next”，



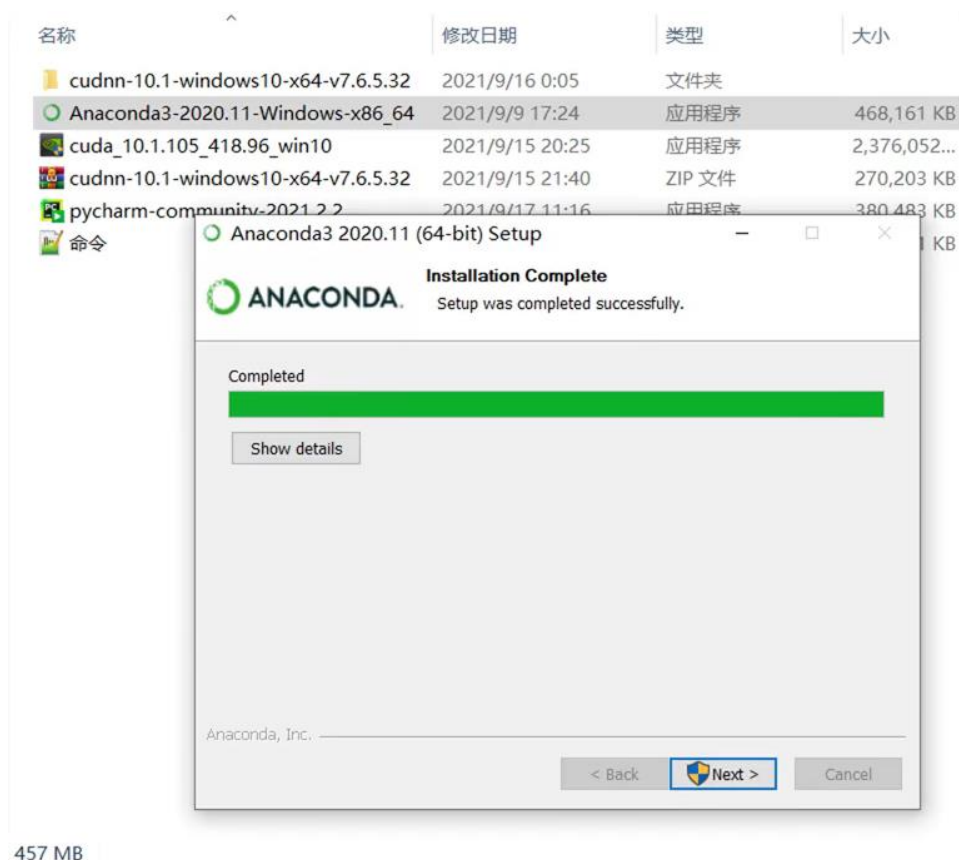
将添加环境变量也勾选上，否则安装完成后还需要自行添加，点击 “Install”，



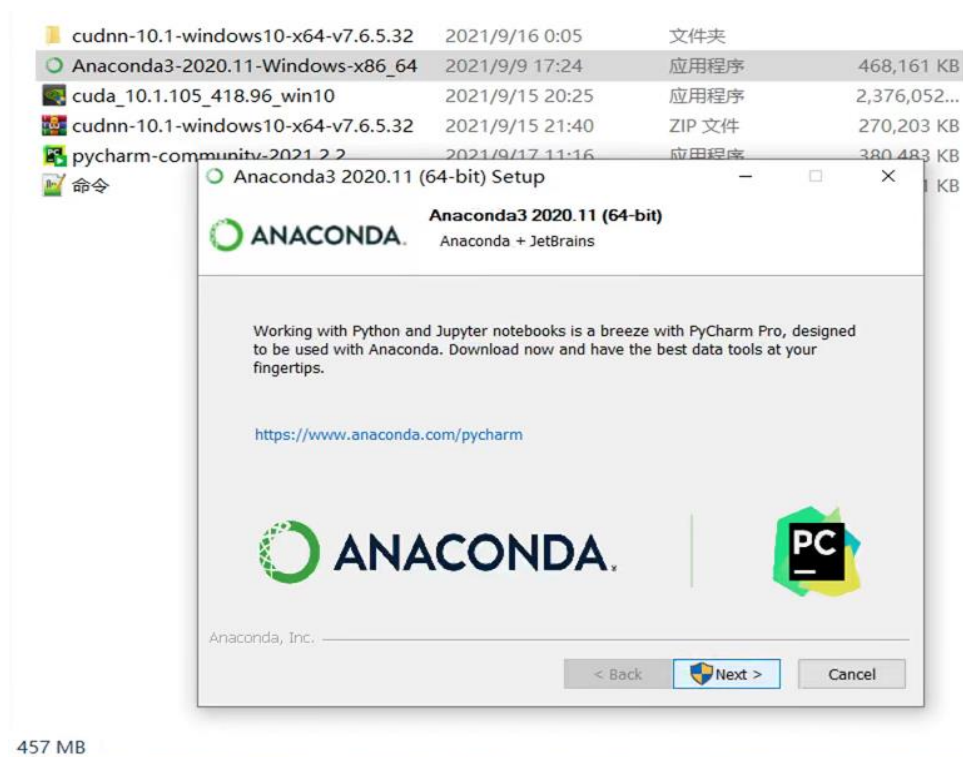
等待安装……



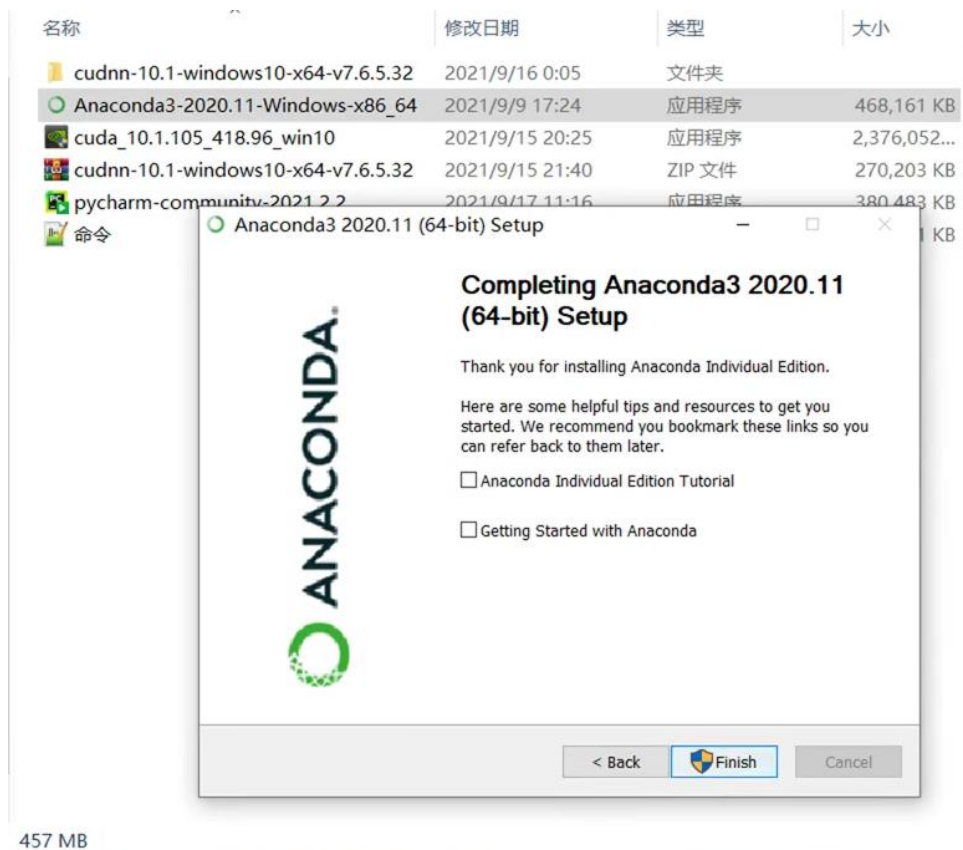
安装完成，点击“Next”，



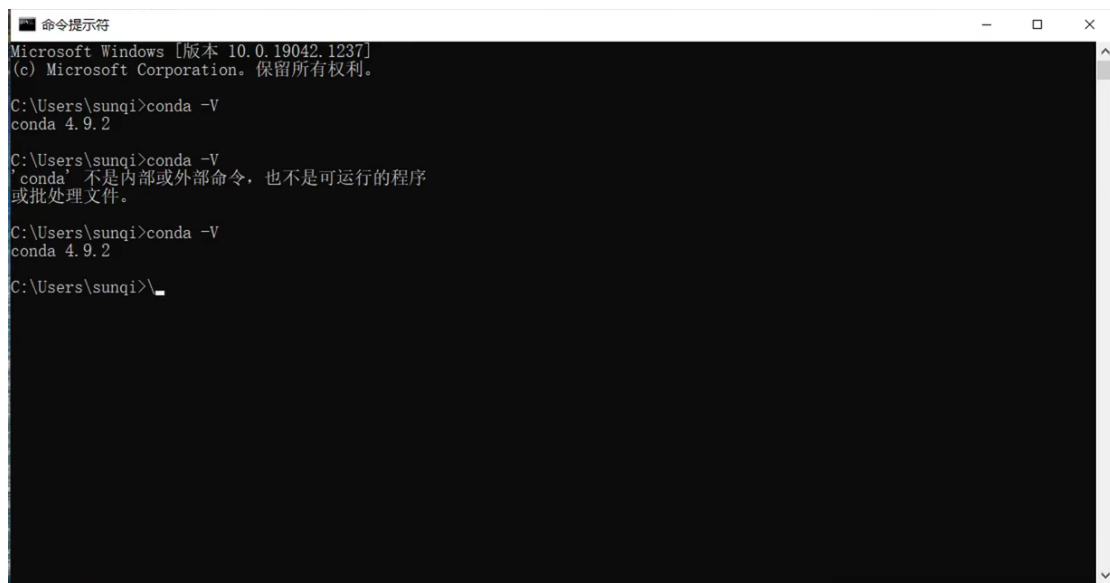
点击“Next”，



两个选框都不勾选，点击“Finish”。



在 cmd 中查看 Anaconda 是否安装成功，输入命令：conda -V



```
命令提示符
Microsoft Windows [版本 10.0.19042.1237]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\sunqi>conda -V
conda 4.9.2

C:\Users\sunqi>conda -V
'conda' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

C:\Users\sunqi>conda -V
conda 4.9.2

C:\Users\sunqi>_
```

Anaconda 安装成功。



```
命令提示符
Microsoft Windows [版本 10.0.19042.1237]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\sunqi>conda -V
conda 4.9.2

C:\Users\sunqi>conda -V
'conda' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

C:\Users\sunqi>conda -V
conda 4.9.2

C:\Users\sunqi>_
```

anaconda安装成功

1.2 Conda 创建虚拟环境

1) 使用 Anaconda 创建虚拟环境

在 cmd 中输入 activate 进入 base 环境，

```
命令提示符
Microsoft Windows [版本 10.0.19042.1237]
(c) Microsoft Corporation。保留所有权利。

C:\Users\sunqi>conda -V
conda 4.9.2

C:\Users\sunqi>conda -V
'conda' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

C:\Users\sunqi>conda -V
conda 4.9.2

C:\Users\sunqi>activate
(base) C:\Users\sunqi>
```

在 base 环境中使用 conda 命令创建环境，下例为创建一个名字为 py37，python 版本为 3.7 的环境，输入命令为：conda create -n py37 python=3.7

```
命令提示符 - conda create -n py37 python=3.7
Microsoft Windows [版本 10.0.19042.1237]
(c) Microsoft Corporation。保留所有权利。

C:\Users\sunqi>conda -V
conda 4.9.2

C:\Users\sunqi>conda -V
'conda' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

C:\Users\sunqi>conda -V
conda 4.9.2

C:\Users\sunqi>activate
(base) C:\Users\sunqi>conda create -n py37 python=3.7
```

输入 y，

【注，若之前已经创建了同样的环境，需要移除掉。】

```
命令提示符 - conda create -n py37 python=3.7
$ conda update -n base -c defaults conda

## Package Plan ##

environment location: C:\Users\sunqi\.conda\envs\py37

added / updated specs:
- python=3.7

The following NEW packages will be INSTALLED:

ca-certificates pkgs/main/win-64::ca-certificates-2021.7.5-haa95532_1
certifi pkgs/main/win-64::certifi-2021.5.30-py37haa95532_0
openssl pkgs/main/win-64::openssl-1.1.1l-h2bbff1b_0
pip pkgs/main/win-64::pip-21.0.1-py37haa95532_0
python pkgs/main/win-64::python-3.7.11-h6244533_0
setuptools pkgs/main/win-64::setuptools-58.0.4-py37haa95532_0
sqlite pkgs/main/win-64::sqlite-3.36.0-h2bbff1b_0
vc pkgs/main/win-64::vc-14.2-h21ff451_1
vs2015_runtime pkgs/main/win-64::vs2015_runtime-14.27.29016-h5e58377_2
wheel pkgs/main/noarch::wheel-0.37.0-pyhd3eb1b0_1
wincertstore pkgs/main/win-64::wincertstore-0.2-py37_0

Proceed ([y]/n)? y
Preparing transaction: /
```

完成安装，

```
命令提示符
done
##
## To activate this environment, use
##
##     $ conda activate py37
##
## To deactivate an active environment, use
##
##     $ conda deactivate
##
(base) C:\Users\sunqi>
```

补充：使用 conda 查看已安装的环境，命令为：conda info -e

2) 使用 Anaconda 激活环境

想要激活环境 py37，在 base 环境中输入：conda activate py37

```
命令提示符
done
##
## To activate this environment, use
##
##     $ conda activate py37
##
## To deactivate an active environment, use
##
##     $ conda deactivate
##
(base) C:\Users\sunqi>condaactivate py37
'condaactivate' 不是内部或外部命令，也不是可运行的程序
或批处理文件。
(base) C:\Users\sunqi>conda activate py37
(py37) C:\Users\sunqi>
```

在环境 py37 中查看 python 的版本，可以看到是 Python 3.7.11 版本，

```
命令提示符 - python
done
##
## To activate this environment, use
##
##     $ conda activate py37
##
## To deactivate an active environment, use
##
##     $ conda deactivate
##
(base) C:\Users\sunqi>condaactivate py37
'condaactivate' 不是内部或外部命令，也不是可运行的程序
或批处理文件。
(base) C:\Users\sunqi>conda activate py37
(py37) C:\Users\sunqi>python
Python 3.7.11 (default, Jul 27 2021, 09:42:29) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

退出 python, 输入: exit()

```
命令提示符
done
#
# To activate this environment, use
#
#     $ conda activate py37
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) C:\Users\sunqi>condaactivate py37
'condaactivate' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

(base) C:\Users\sunqi>conda activate py37

(py37) C:\Users\sunqi>python
Python 3.7.11 (default, Jul 27 2021, 09:42:29) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()

(py37) C:\Users\sunqi>
```

3) 使用 Anaconda 退出环境

想要退出环境 py37, 在 py37 环境中输入: conda deactivate

```
命令提示符 - conda deactivate
done
#
# To activate this environment, use
#
#     $ conda activate py37
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) C:\Users\sunqi>condaactivate py37
'condaactivate' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

(base) C:\Users\sunqi>conda activate py37

(py37) C:\Users\sunqi>python
Python 3.7.11 (default, Jul 27 2021, 09:42:29) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()

(py37) C:\Users\sunqi>conda deactivate

(base) C:\Users\sunqi>
```

查看 base 环境中的 python 版本,

```
命令提示符 - conda deactivate - python
done
#
# To activate this environment, use
#
#     $ conda activate py37
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) C:\Users\sunqi>condaactivate py37
'condaactivate' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

(base) C:\Users\sunqi>conda activate py37

(py37) C:\Users\sunqi>python
Python 3.7.11 (default, Jul 27 2021, 09:42:29) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()

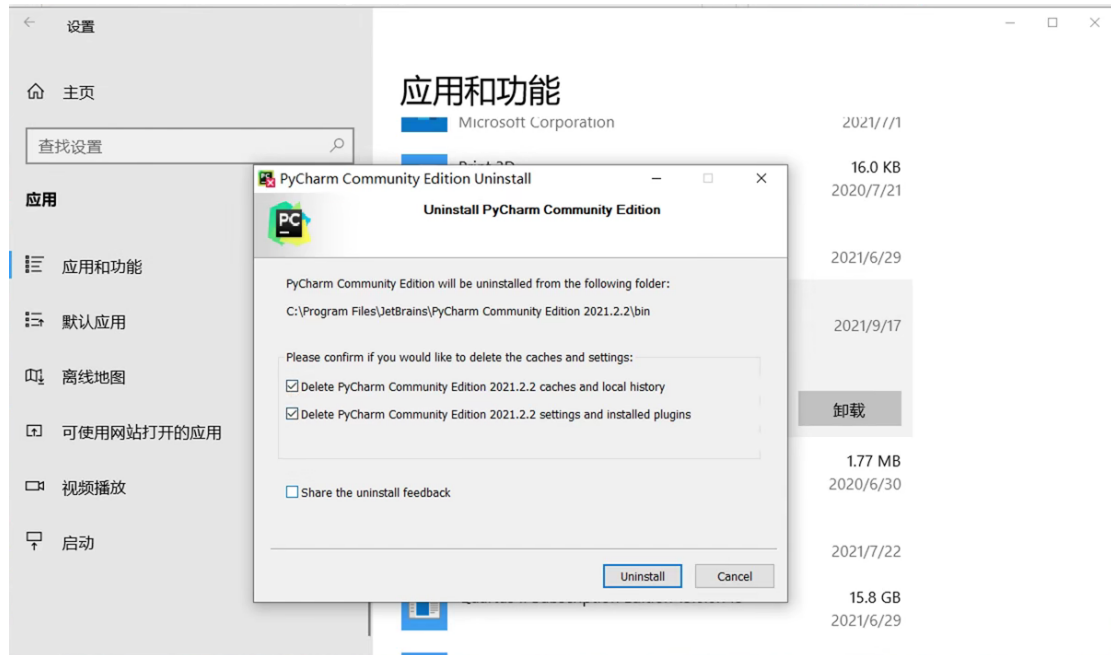
(py37) C:\Users\sunqi>conda deactivate

(base) C:\Users\sunqi>python
Python 3.8.5 (default, Sep 3 2020, 21:29:08) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

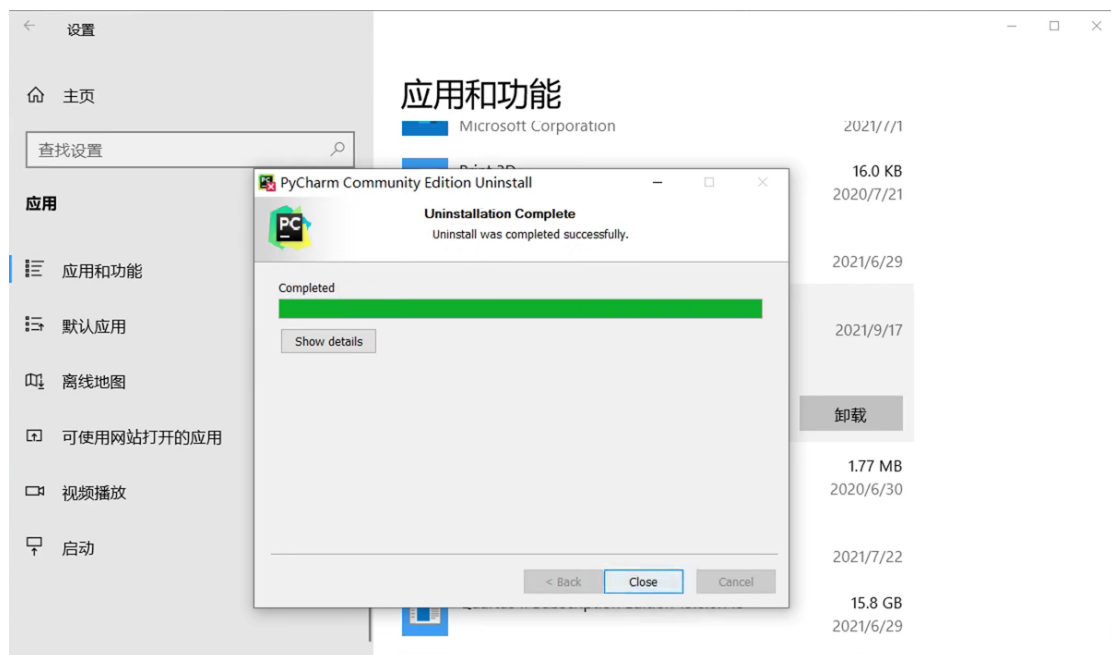
1.3 PyCharm 的安装和环境、解释器配置

1) PyCharm 的卸载

若之前已安装 PyCharm，想要卸载后重装，则需要点击“卸载”后勾选两个方框，以便卸载干净，点击“Uninstall”，



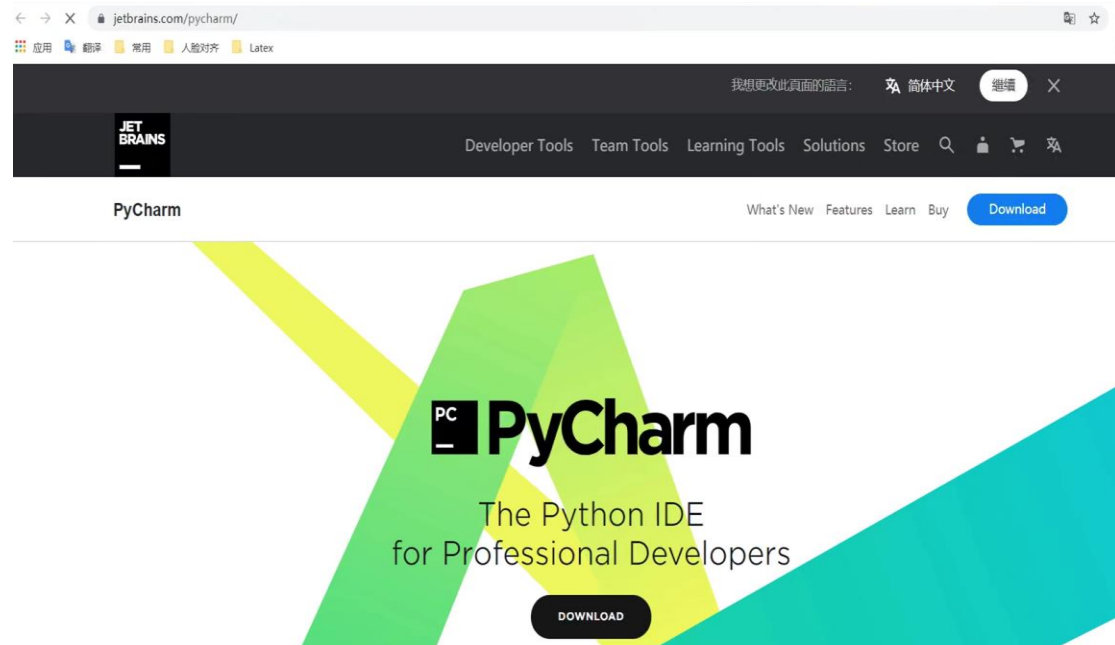
卸载完成，点击“Close”，



2) PyCharm 的安装

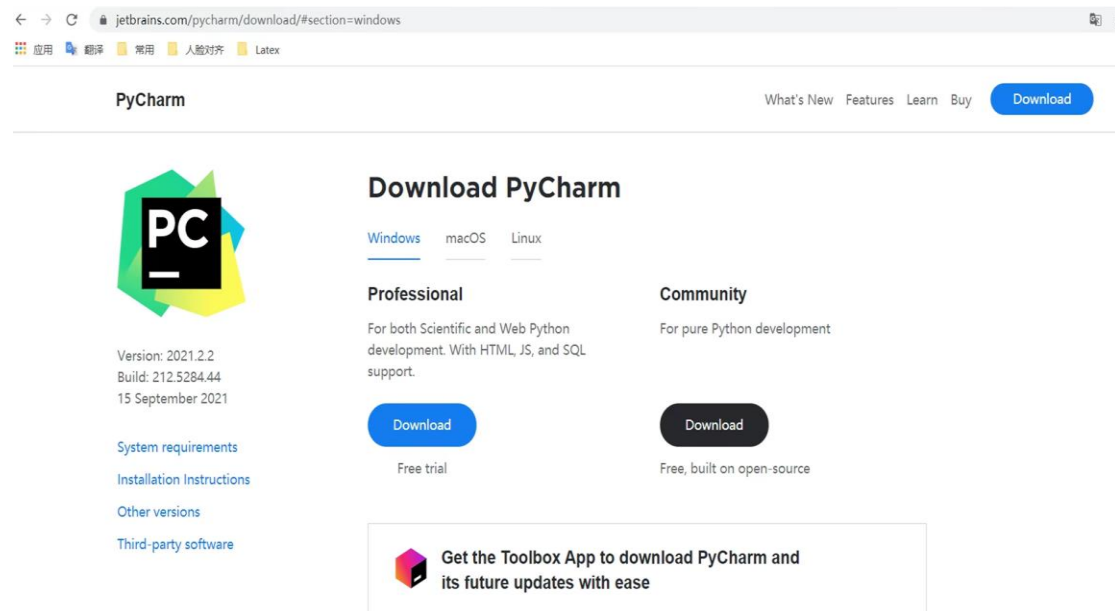
PyCharm 官网地址: <https://www.jetbrains.com/pycharm/>

点击“DOWNLOAD”，



选择“Community”版本，

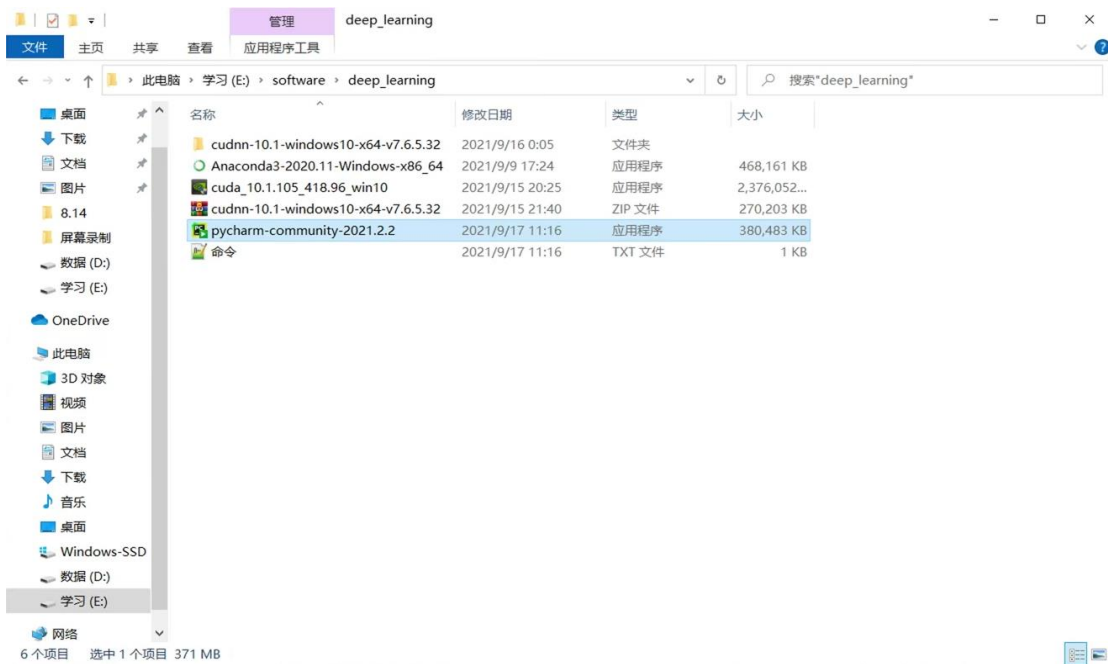
【注：经过官网的学生认证，在校大学生可以申请免费使用“Professional”版本。】



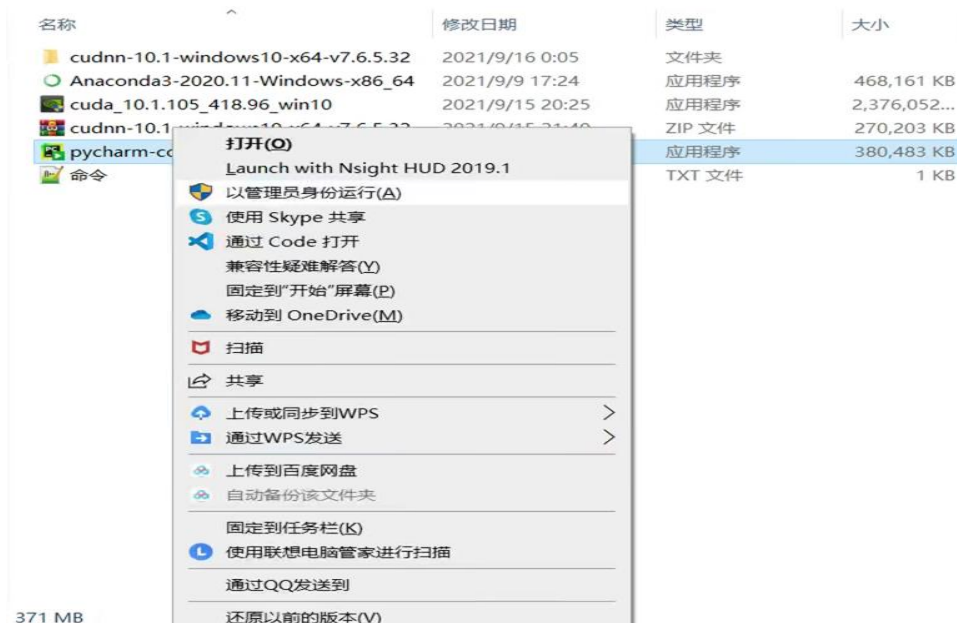
下载“Community”版本中……



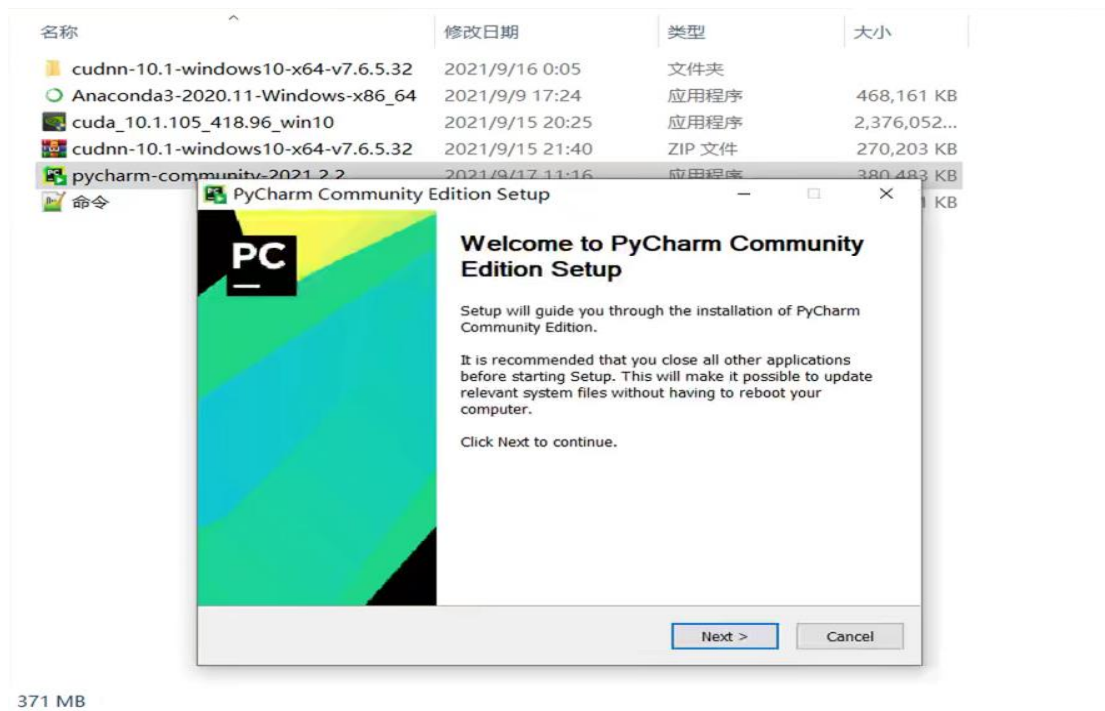
下载成功，存放在本地硬盘中，如



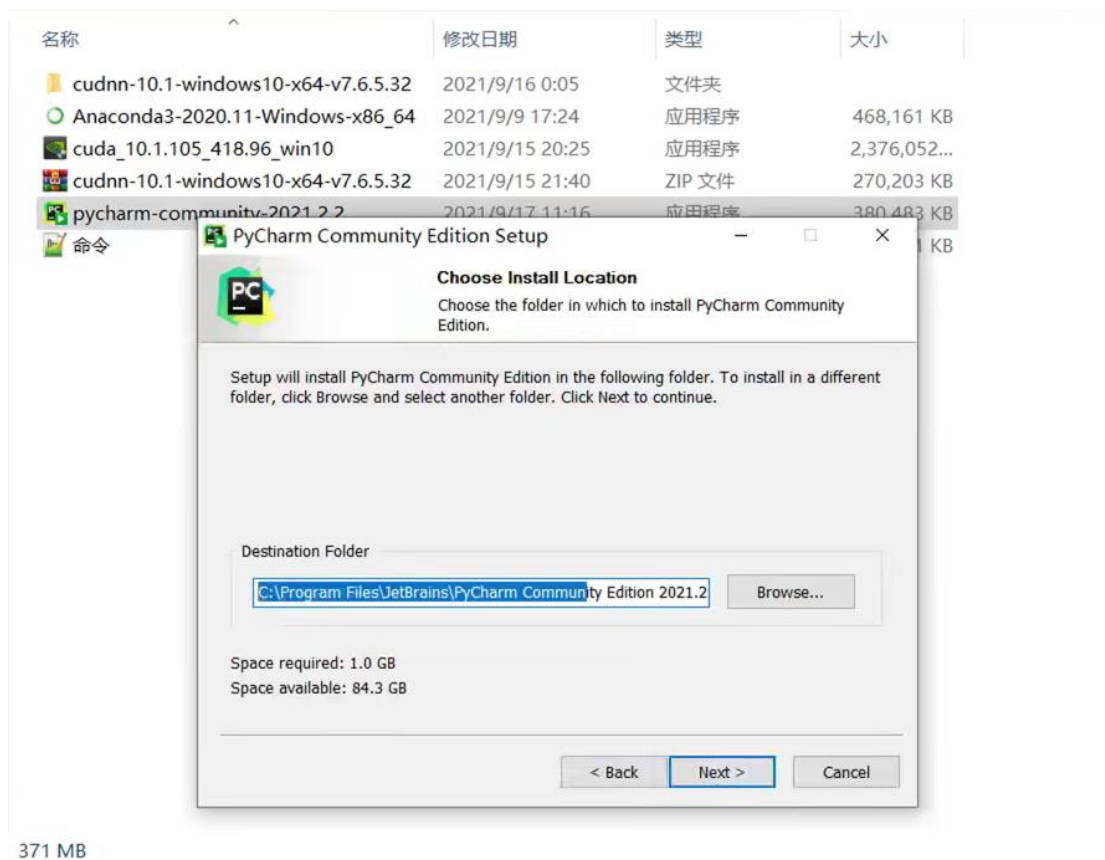
右键“pycharm-community-2021.2.2”应用程序，选择“以管理员身份运行”，



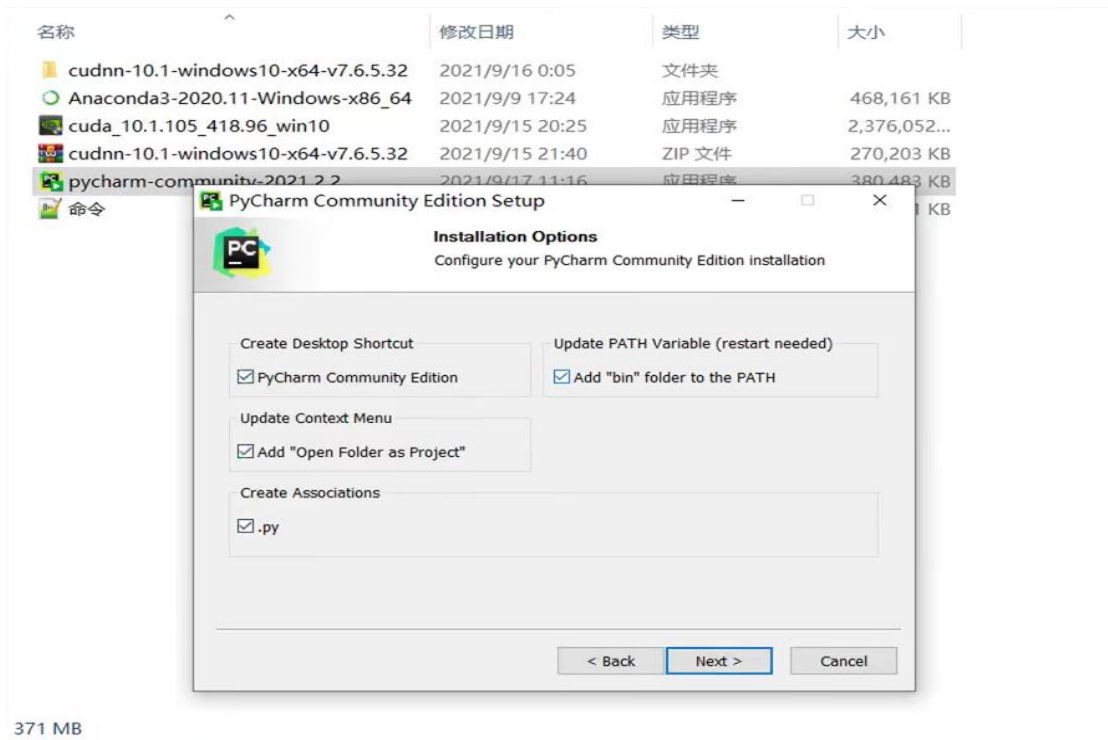
点击“Next”，



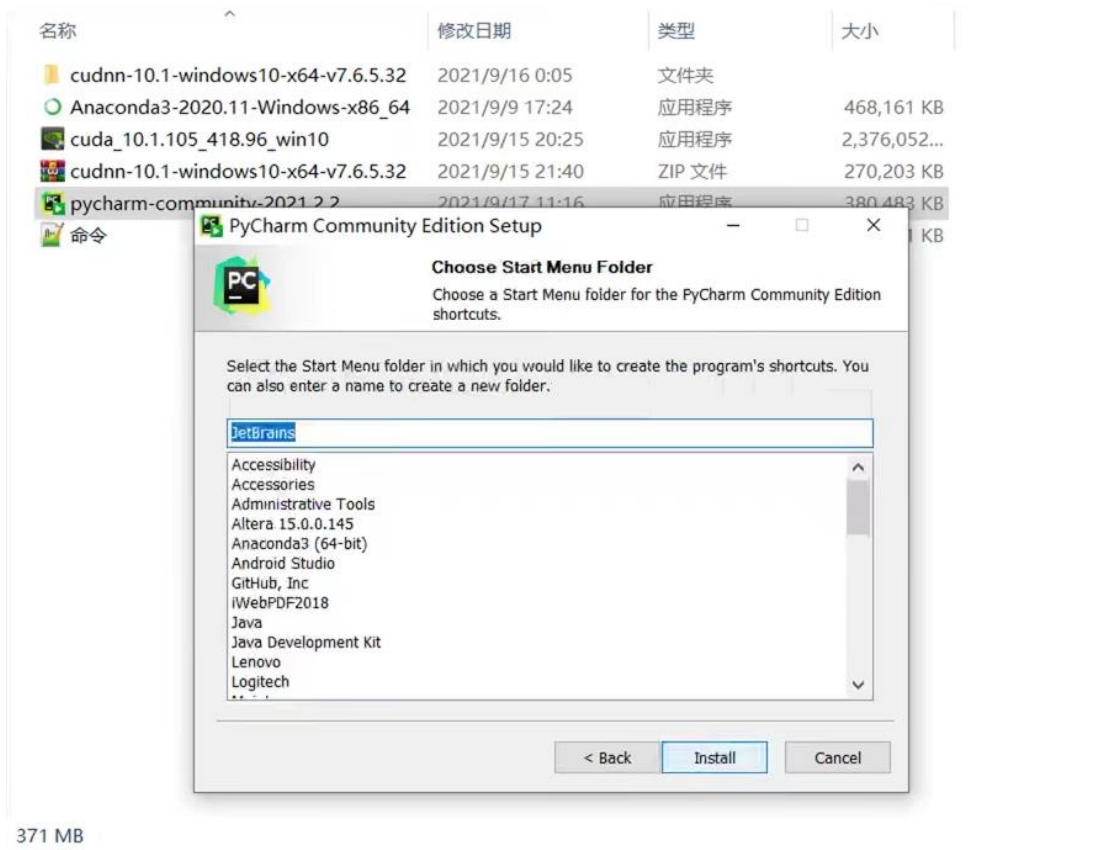
可以修改安装路径，可以默认为 C 盘或者改为 D 盘，点击“Next”，



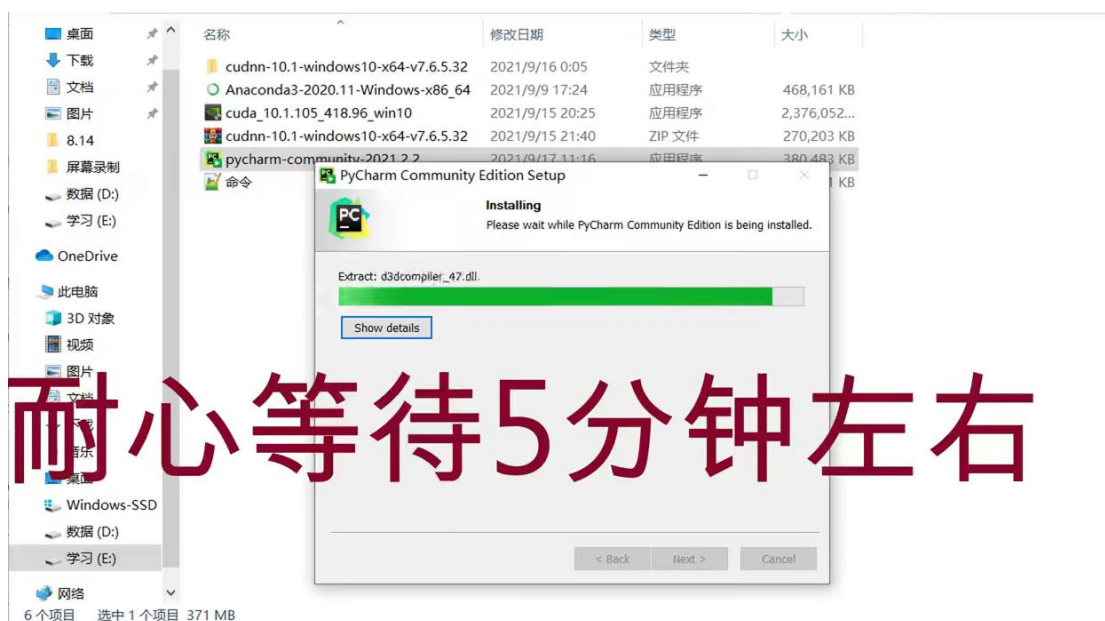
建议勾选上所有选框，点击“Next”，



默认，不要改变，点击“Install”，

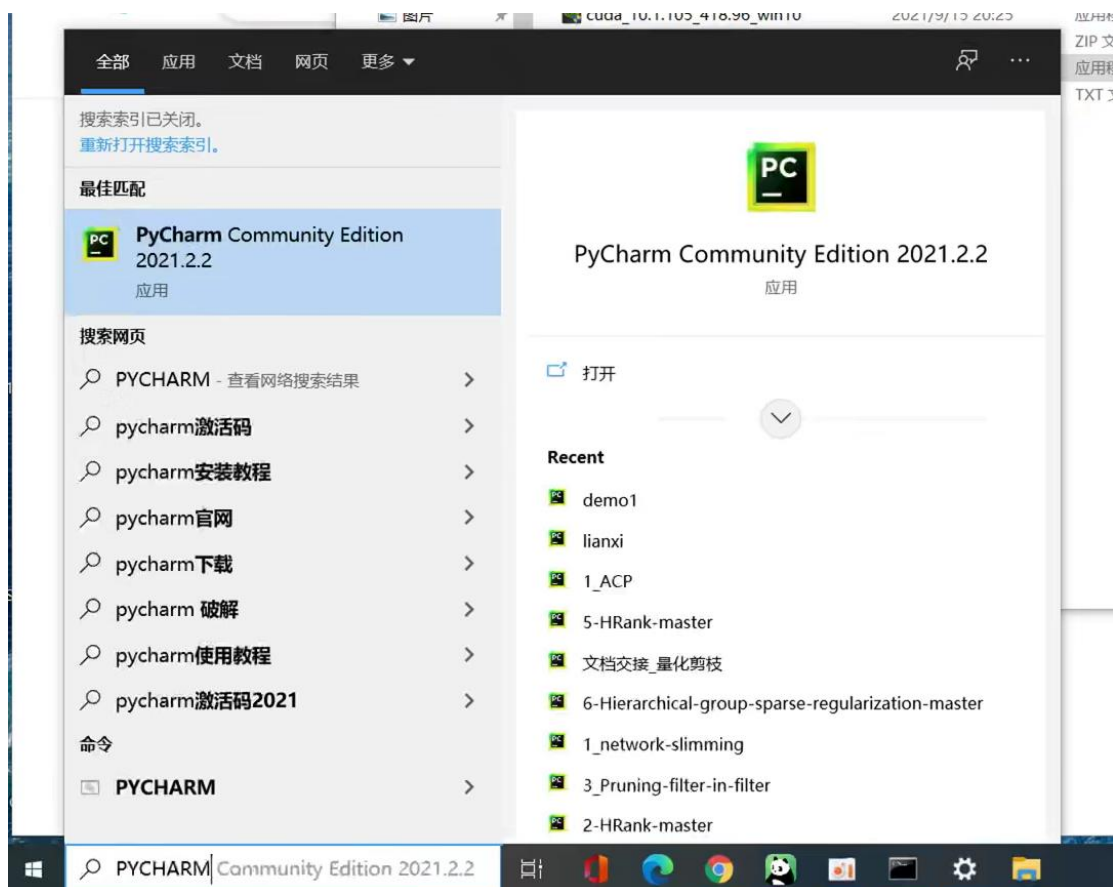


安装中……



选择待会儿重启，点击“Finish”即可。

打开 PyCharm，

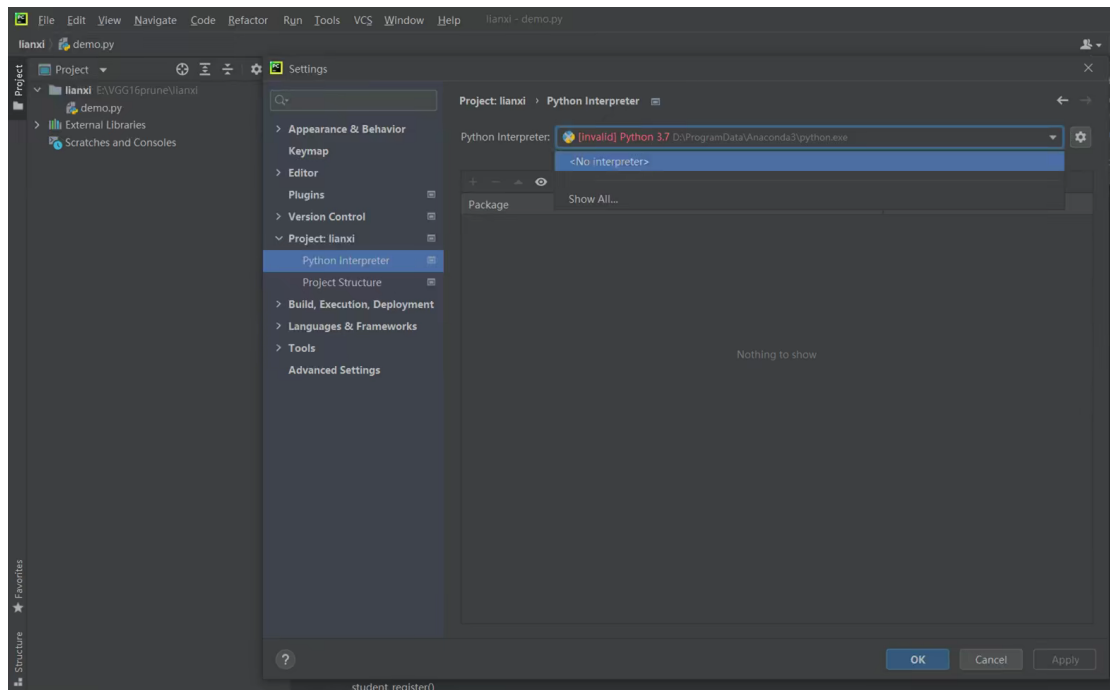


打开界面，

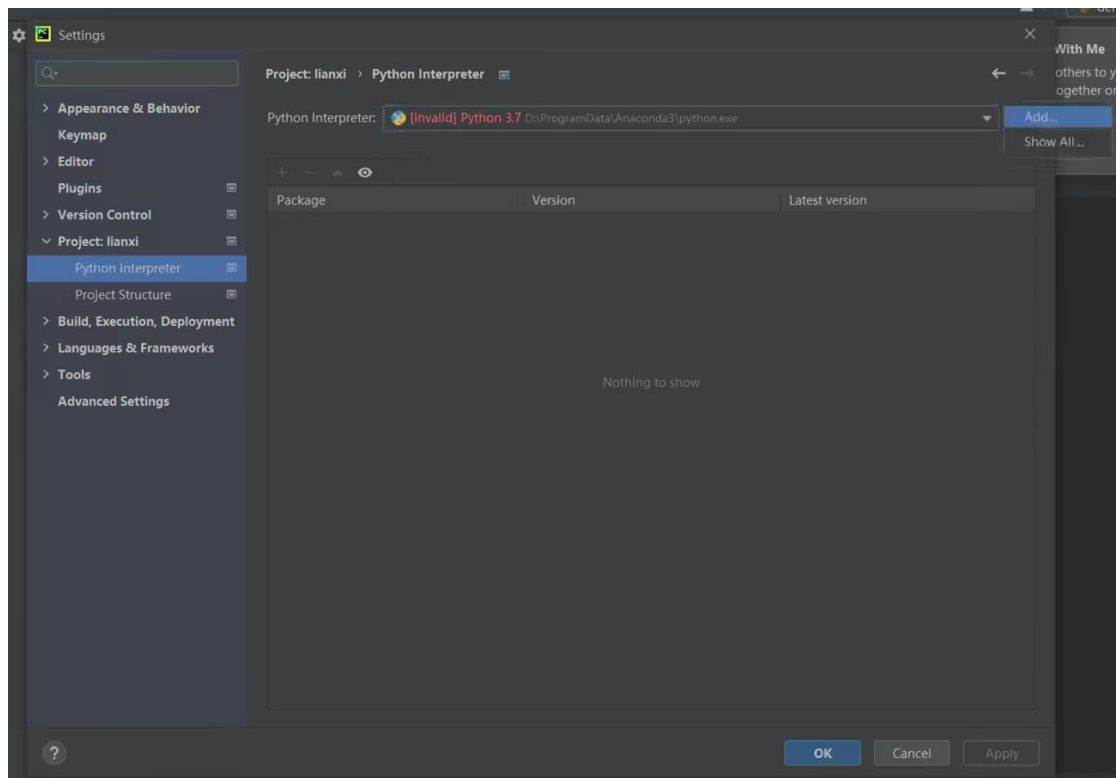


3) PyCharm 的解释器配置

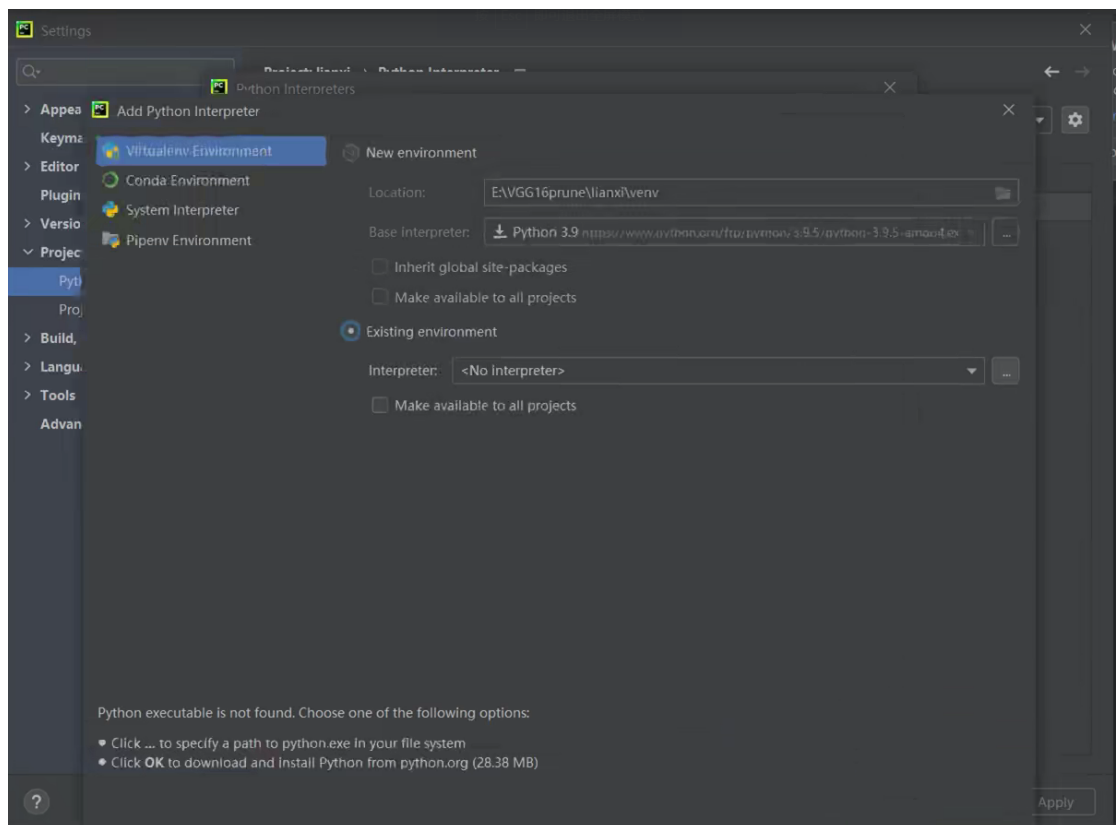
PyCharm 打开代码运行时，需要设置解释器，依次点击 File-Settings-Project lianxi-IPython Interpreter



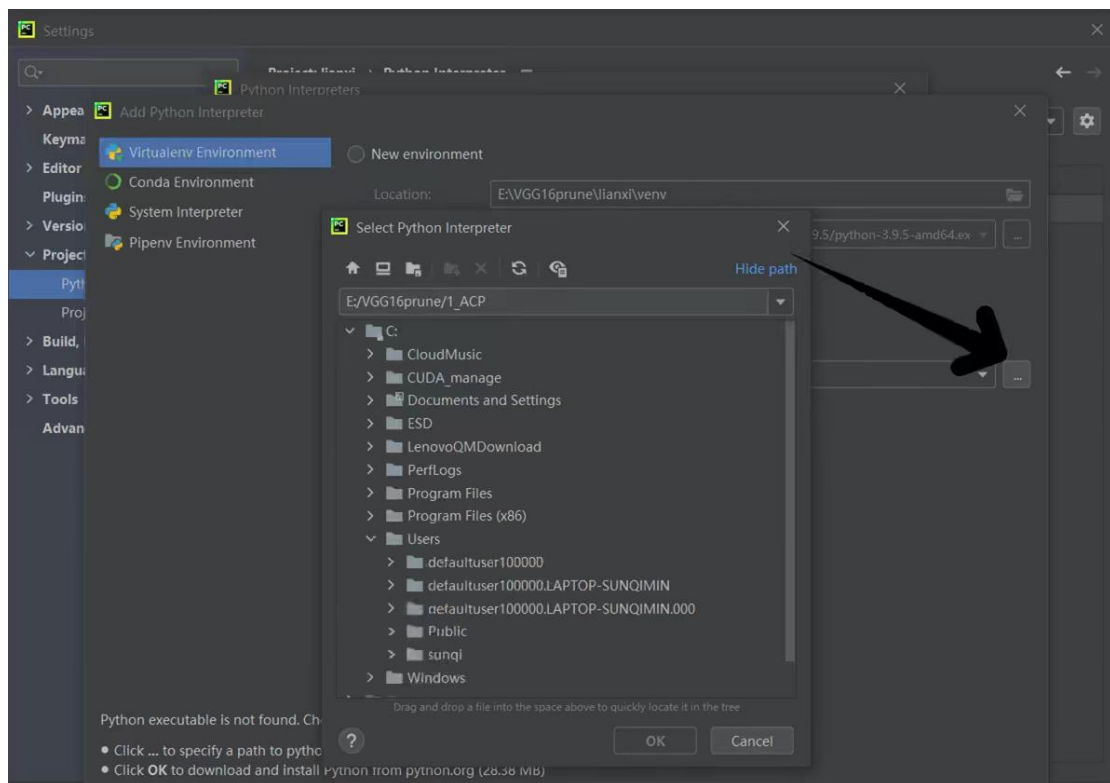
点击右侧齿轮，选择“Add...”，



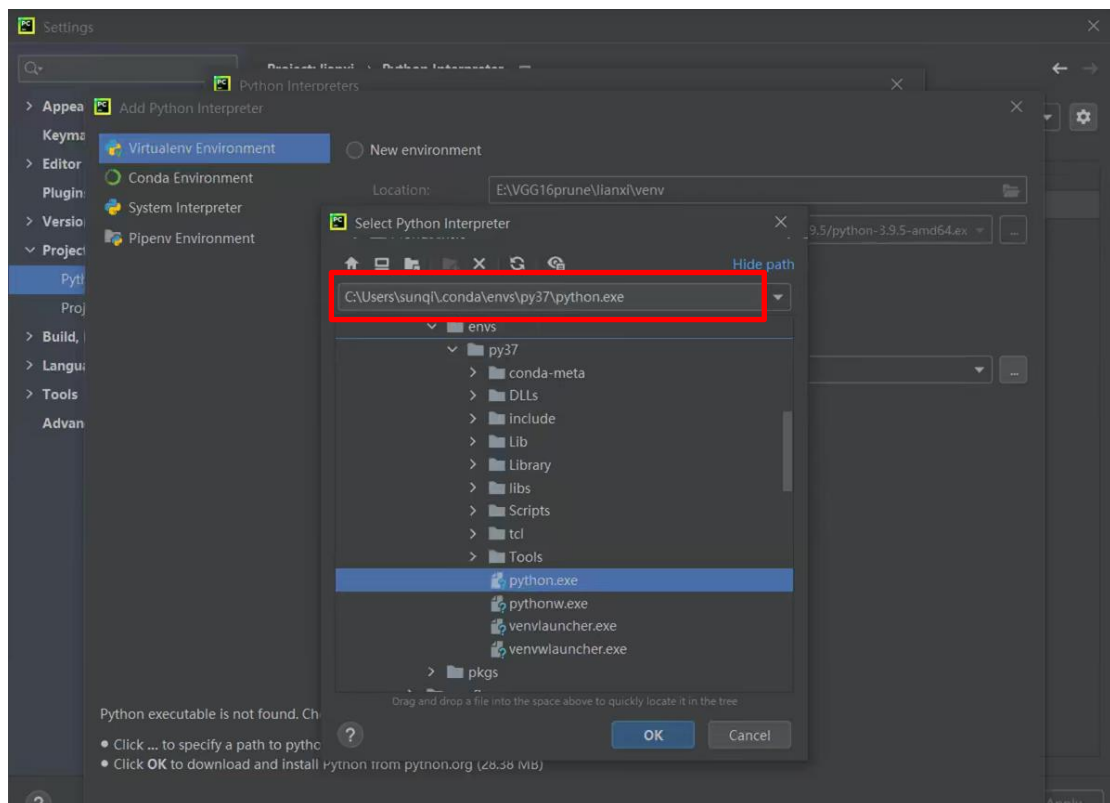
选择“Virtualenv Environment”中已存在的环境，



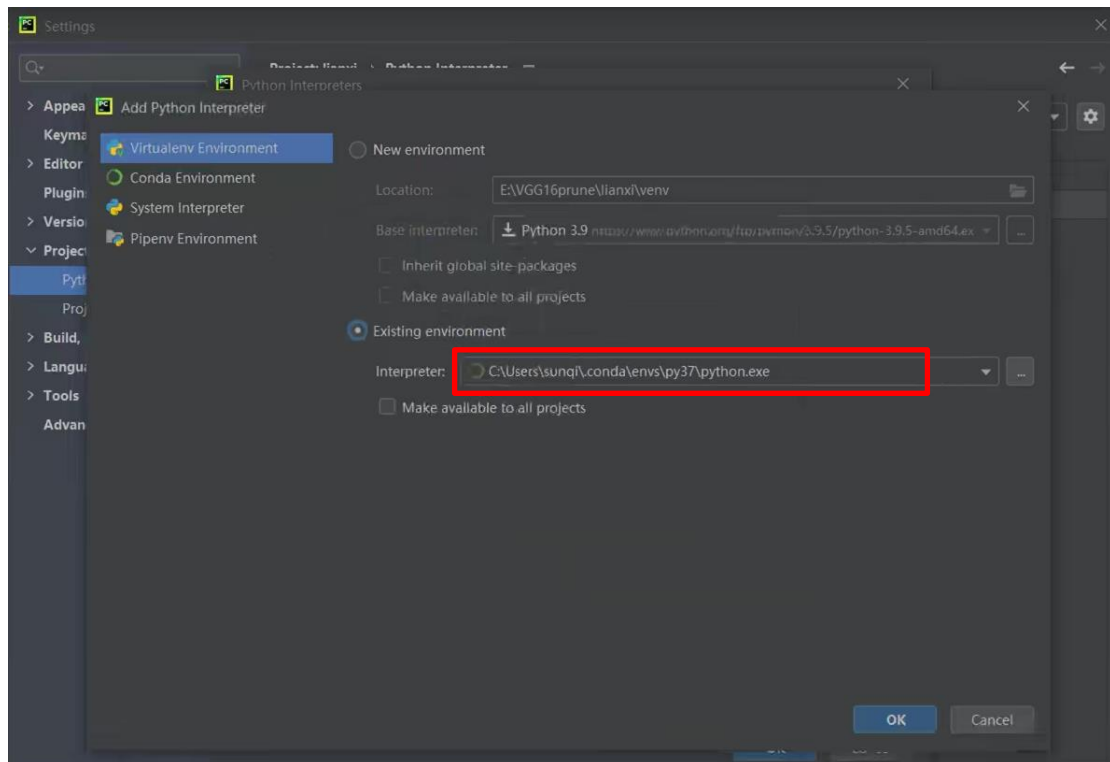
点击右侧“...”，



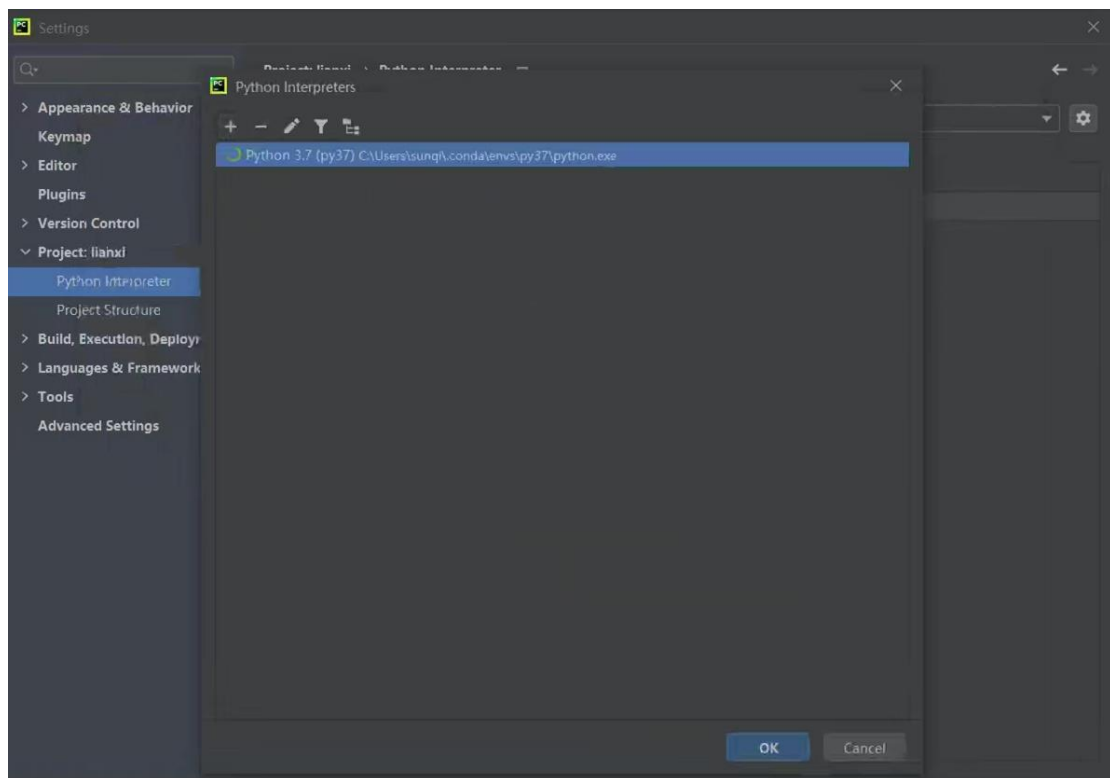
找到之前创建的环境 py37 中的 python.exe 所在位置，



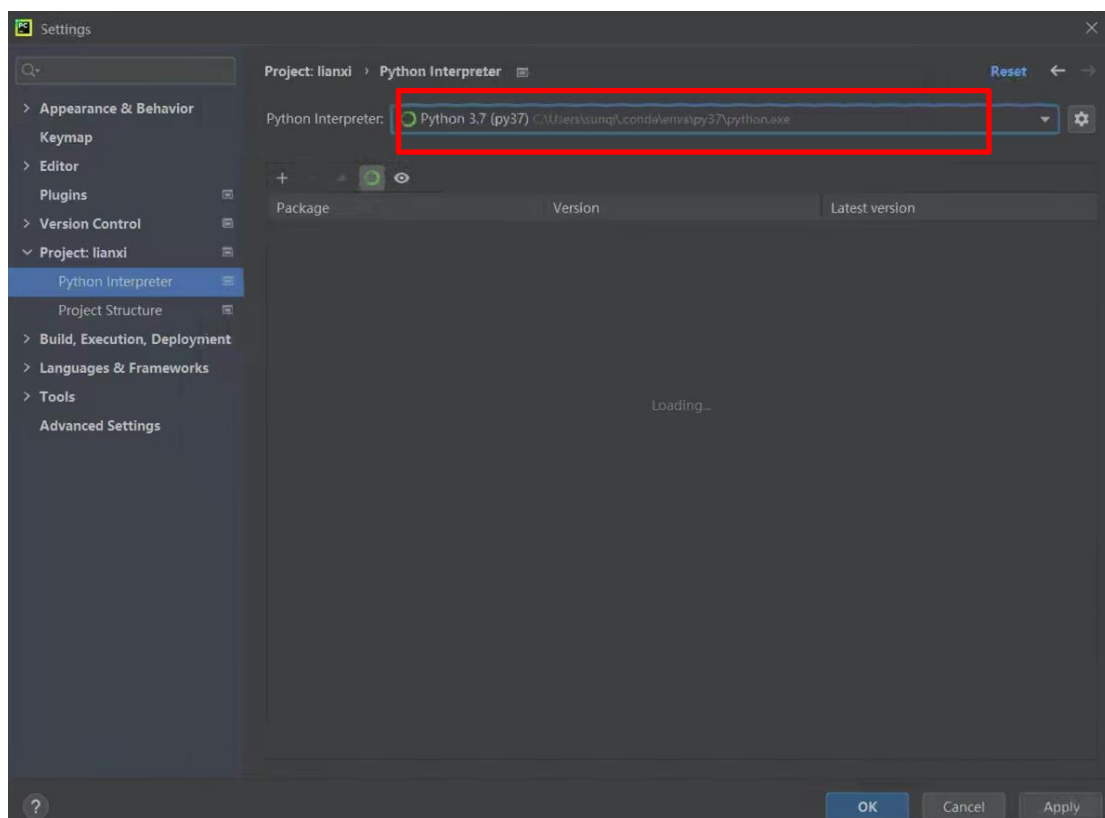
双击或者点击“OK”，



点击“OK”，



点击“ok”，

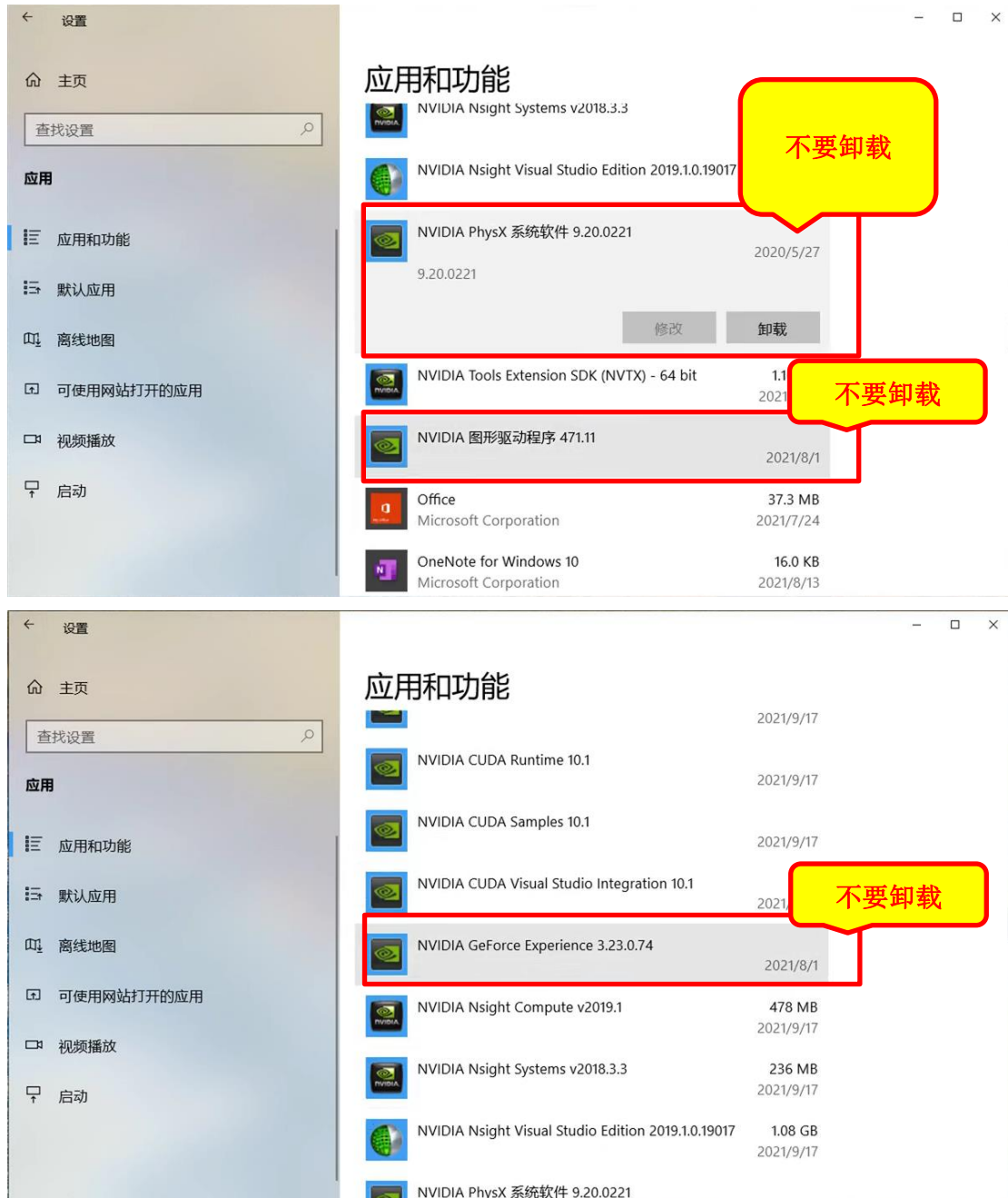


点击“ok”。

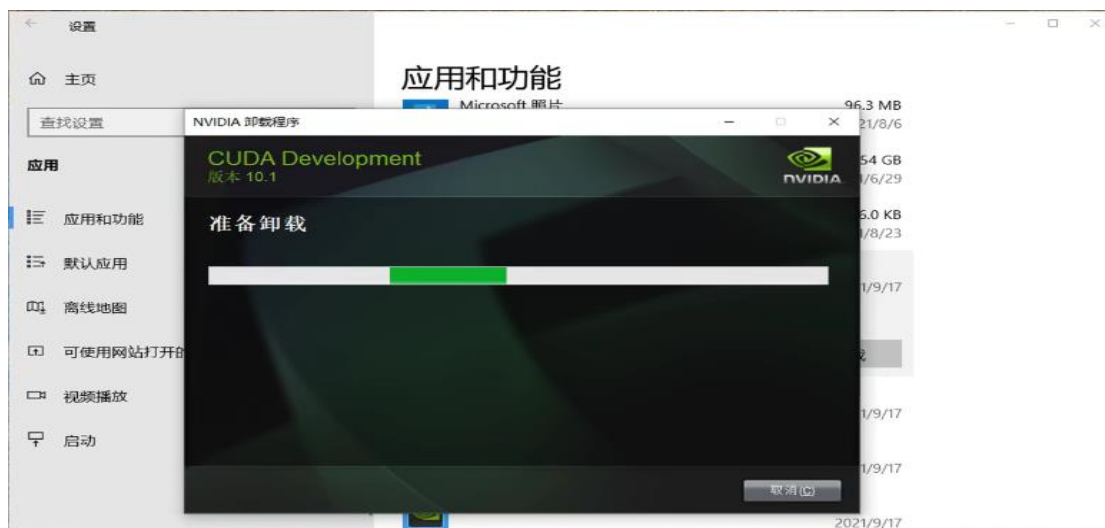
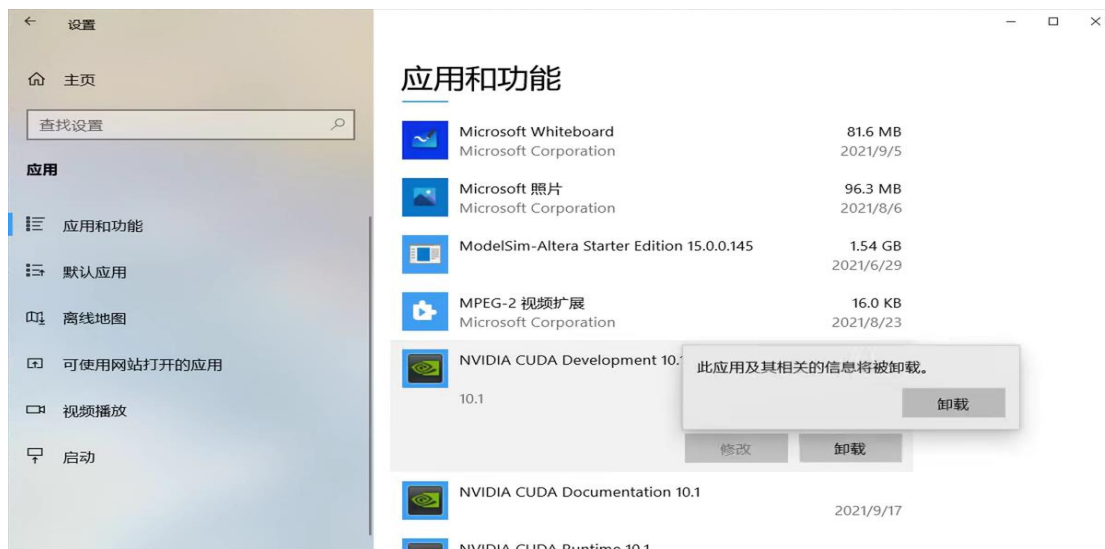
1.4 CUDA 的安装

1) CUDA 的卸载

若之前已安装 CUDA, 想要卸载后重装, 则需要点击“卸载”, 此时, 需要保留 NVIDIA PhysX 系统软件、NVIDIA 图形驱动程序、NVIDIA GeForce Experience, 除了这三个, 其余的 NVIDIA 的均可卸载。



卸载 CUDA,



卸载 CUDA Documentation,



卸载 CUDA Samples 等,



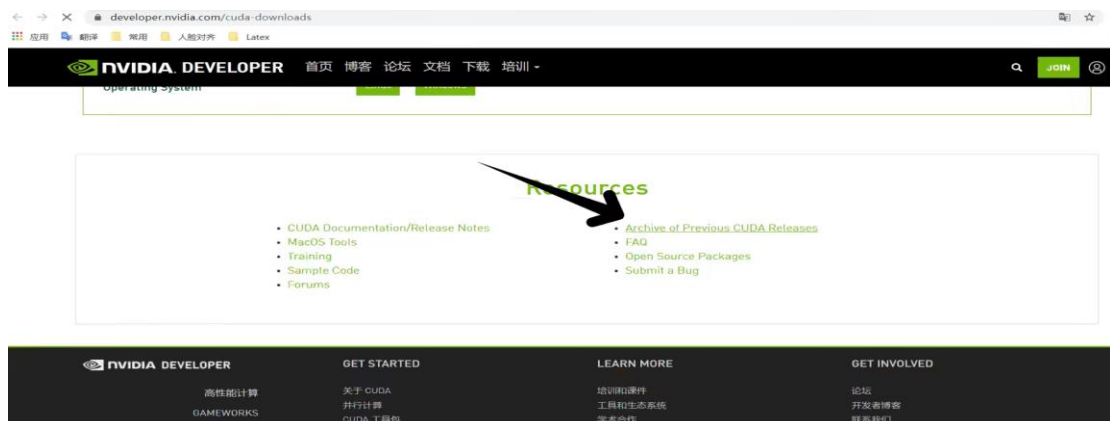
卸载完成，保留了 NVIDIA PhysX 系统软件、NVIDIA 图形驱动程序、NVIDIA GeForce Experience。



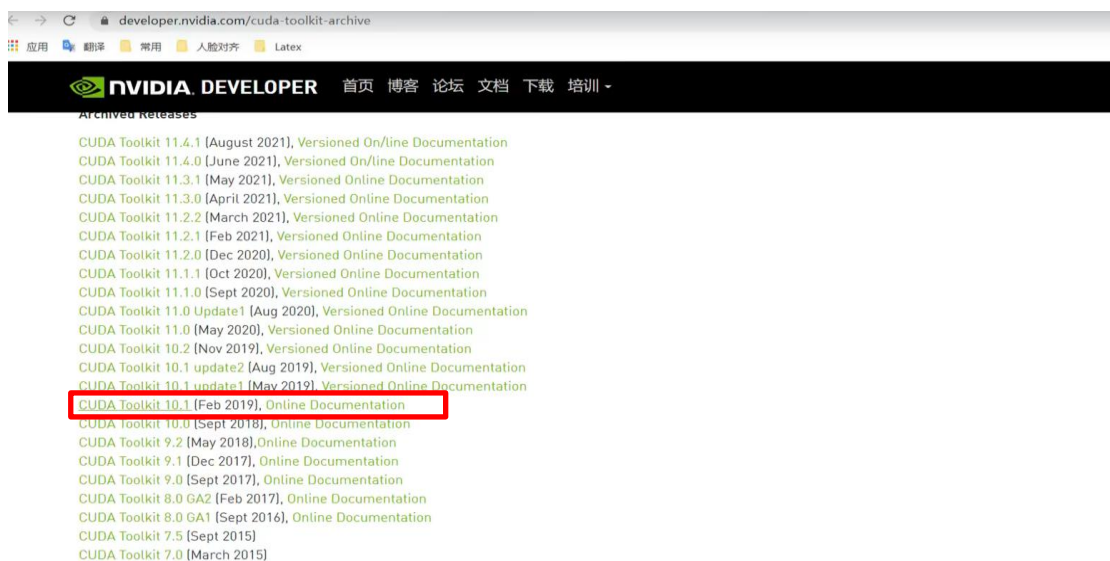
2) CUDA 的安装

CUDA 官网: <https://developer.nvidia.com/cuda-downloads>

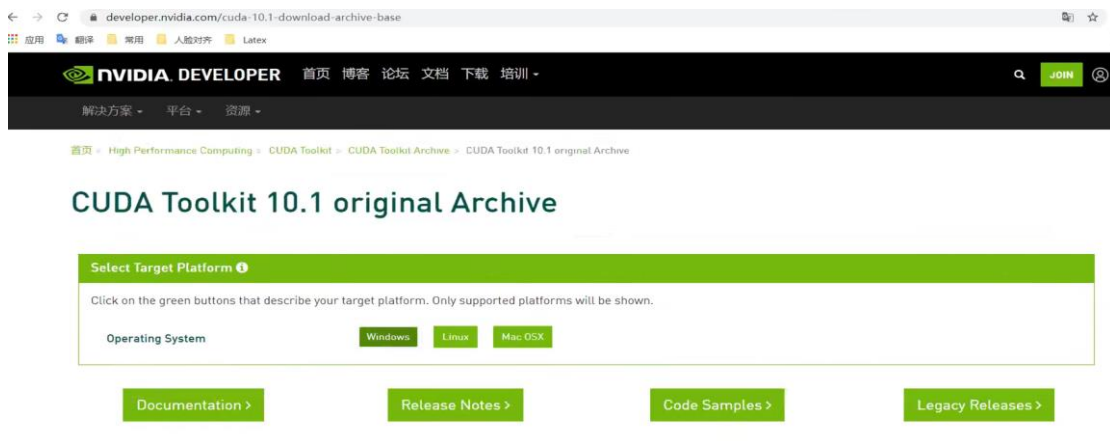
打开 CUDA 官网，选择之前的版本，



根据已安装 conda4.9.2 的版本，选择 CUDA 版本，如，

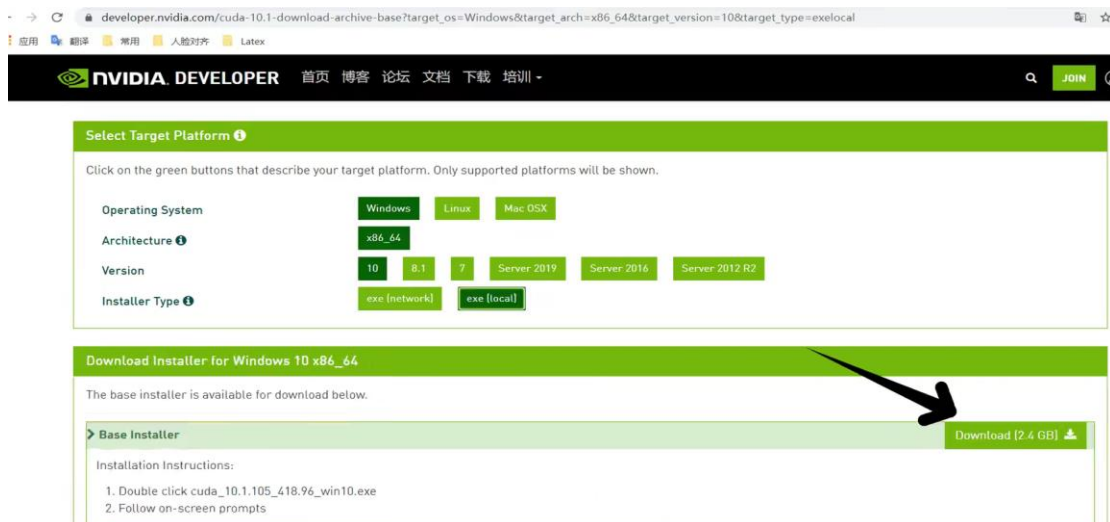


点击“CDUA Toolkit 10.1”，

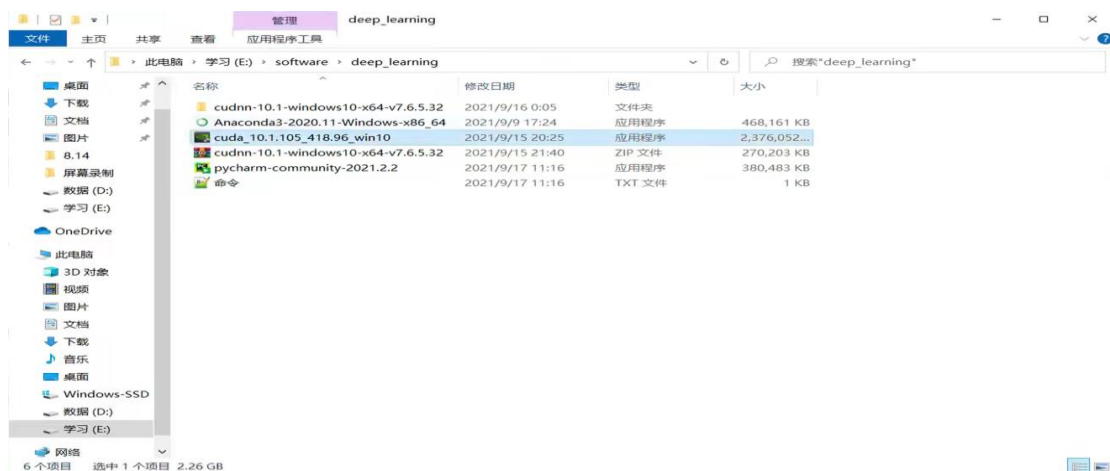


Additional Resources

选择版本，如 Windows-x86_64-10-exe[local]，点击“Download”，



下载完成，存放到本地硬盘中，

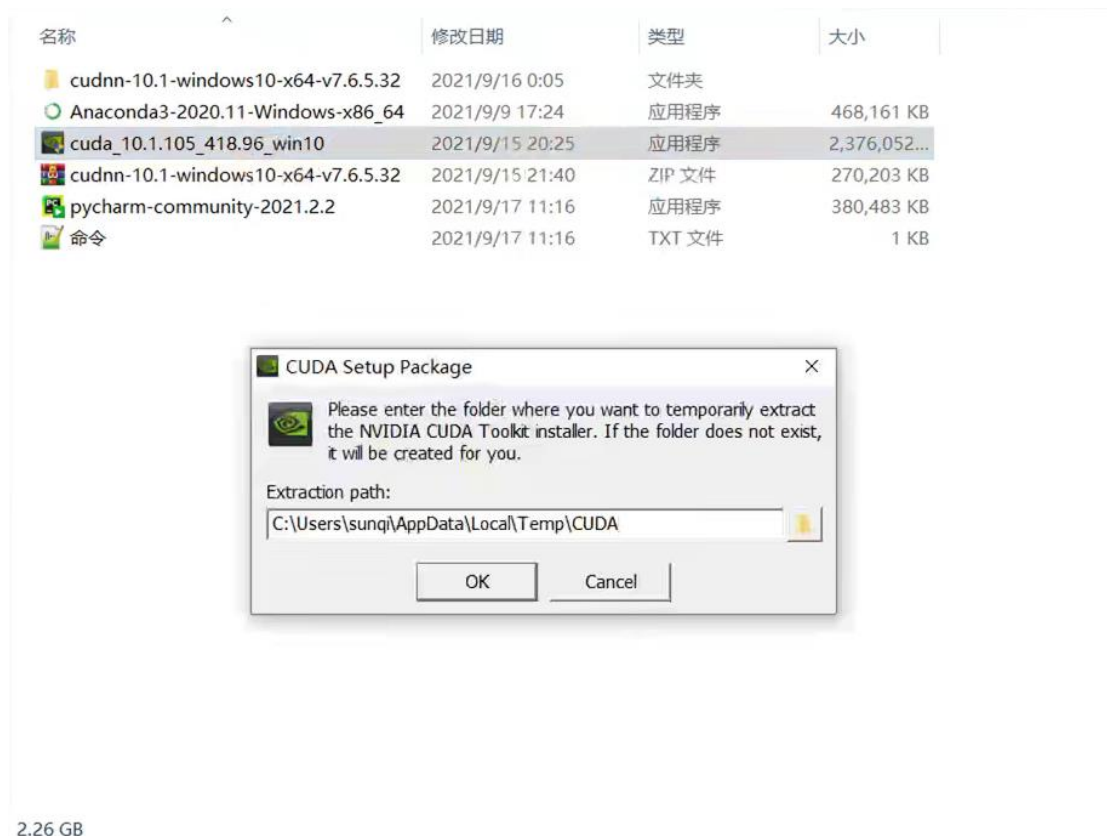


右键“cuda_10.1.105_418.96_win10”应用程序，选择“以管理员身份运行”，

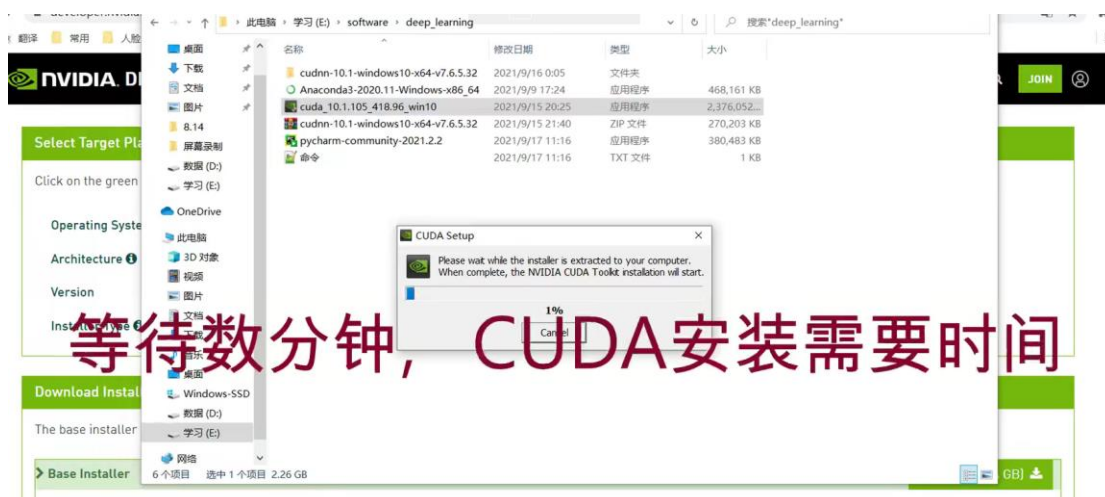


点击“OK”，

【注，此处不太建议修改安装路径。】



安装中.....



点击“同意并继续”，



选择“自定义”，点击“下一步”，



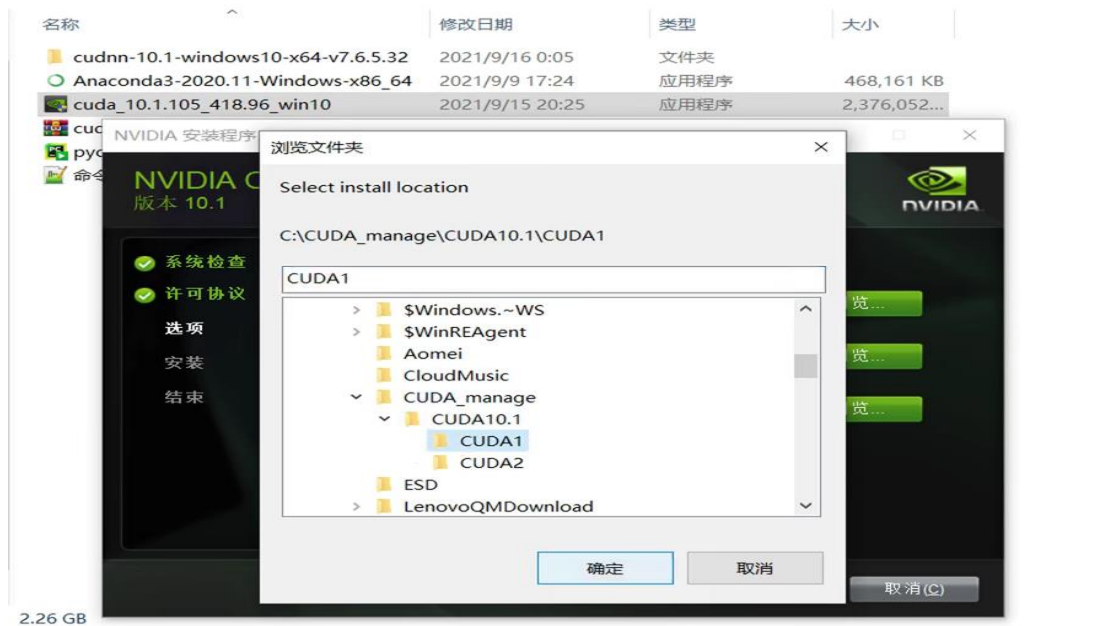
自定义安装选项中，需要注意，新版本要比旧版本高才勾选，如本例中，之前保留的版本比新版本高，就不勾选。点击“下一步”，



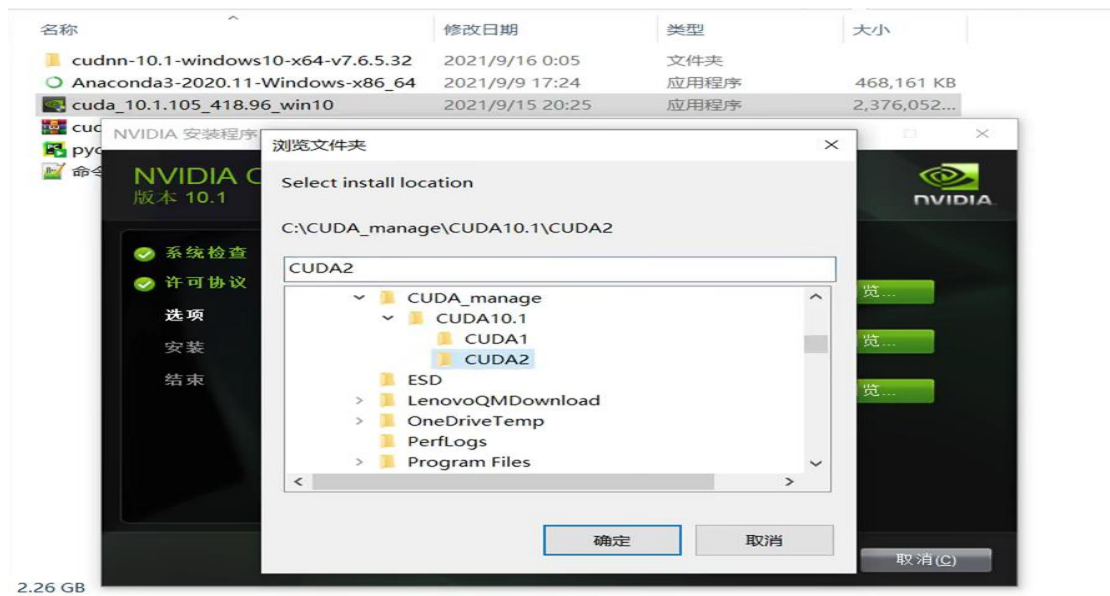
选择安装位置，建议在 C 盘或者 D 盘建立文件夹，以便不同的 CUDA 版本使用。如在 C 盘，新建了文件夹“CUDA_manage”，之后新建文件夹“CUDA10.1”，然后新建两个文件夹“CUDA1”和“CUDA2”，



前两个选择“CUDA1”，



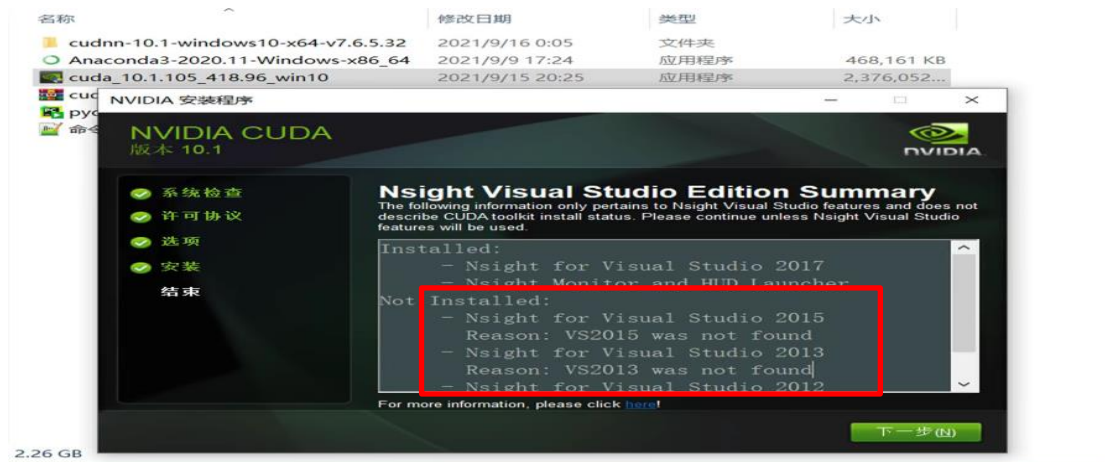
第三个选择“CUDA2”，



点击“下一步”，等待安装，



安装中出现的提示，本地中未安装 Visual Studio，可以自行安装，不装也不影响，点击“下一步”，



下面两个框均不勾选，点击“关闭”。



查看 CUDA 是否安装成功，输入命令：nvcc -V

```
命令提示符 - conda deactivate
# To deactivate an active environment, use
#
# $ conda deactivate

(base) C:\Users\sunqi>condaactivate py37
'condaactivate' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

(base) C:\Users\sunqi>conda activate py37

(py37) C:\Users\sunqi>python
Python 3.7.11 (default, Jul 27 2021, 09:42:29) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()

(py37) C:\Users\sunqi>conda deactivate

(base) C:\Users\sunqi>python
Python 3.8.5 (default, Sep 3 2020, 21:29:08) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()

(base) C:\Users\sunqi>nvcc -V
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2019 NVIDIA Corporation
Built on Fri_Feb_8_19:08:26_Pacific_Standard_Time_2019
Cuda compilation tools, release 10.1, V10.1.105

(base) C:\Users\sunqi>
```

1.5 cuDNN 神经网络库的配置

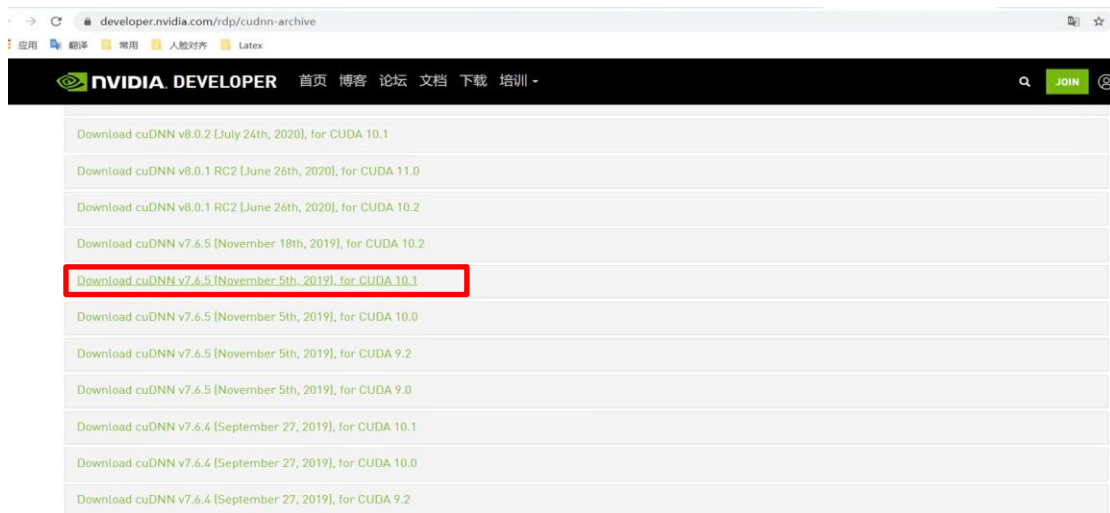
cuDNN 官网: <https://developer.nvidia.com/zh-cn/cudnn>

打开官网



Up to 5x Faster with cuDNN 8.0(A100) vs. cuDNN 7.6 (V100)

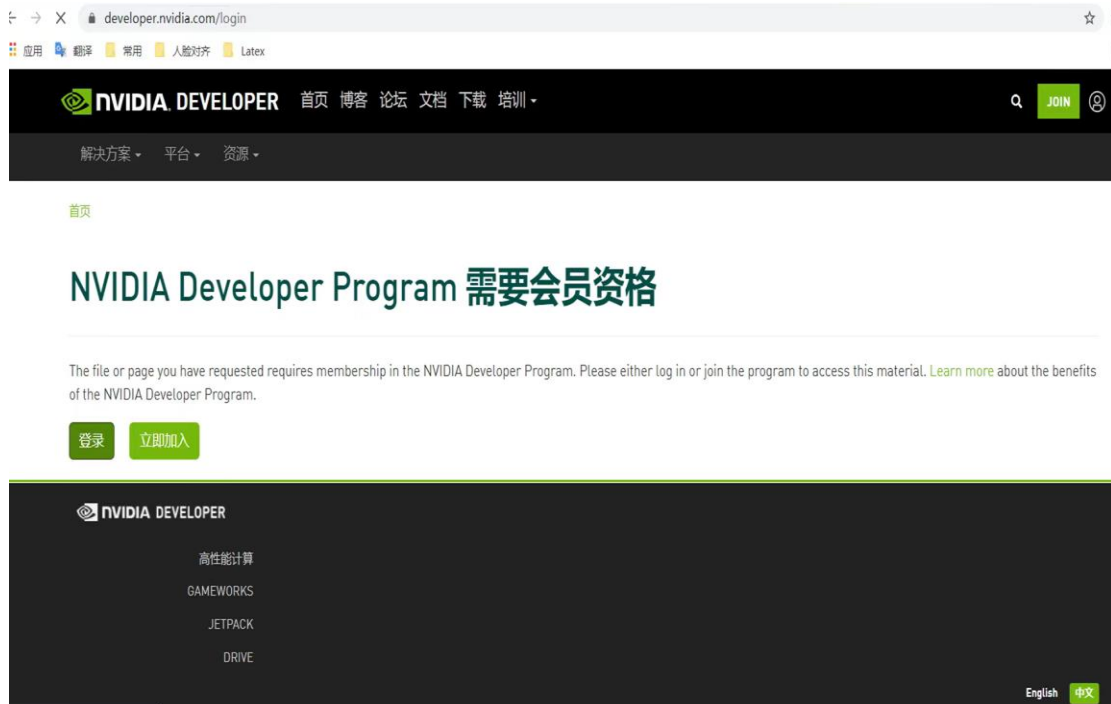
选择与安装的 CUDA 对应的版本,



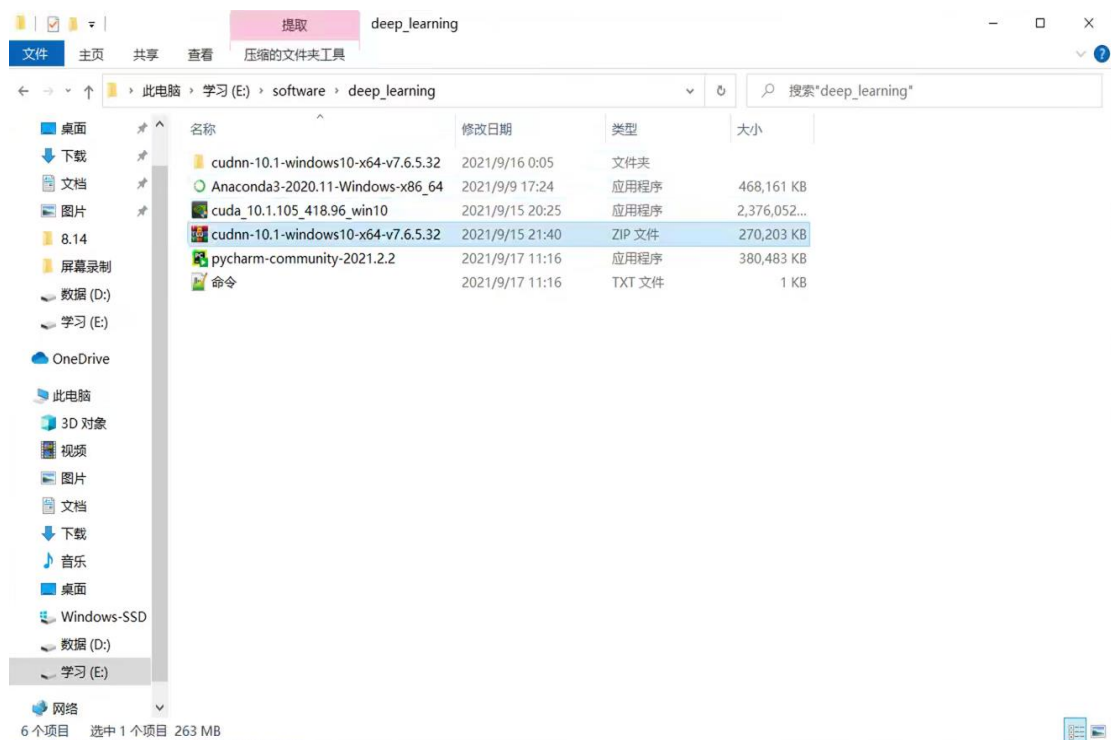
选择本机操作系统匹配的版本,



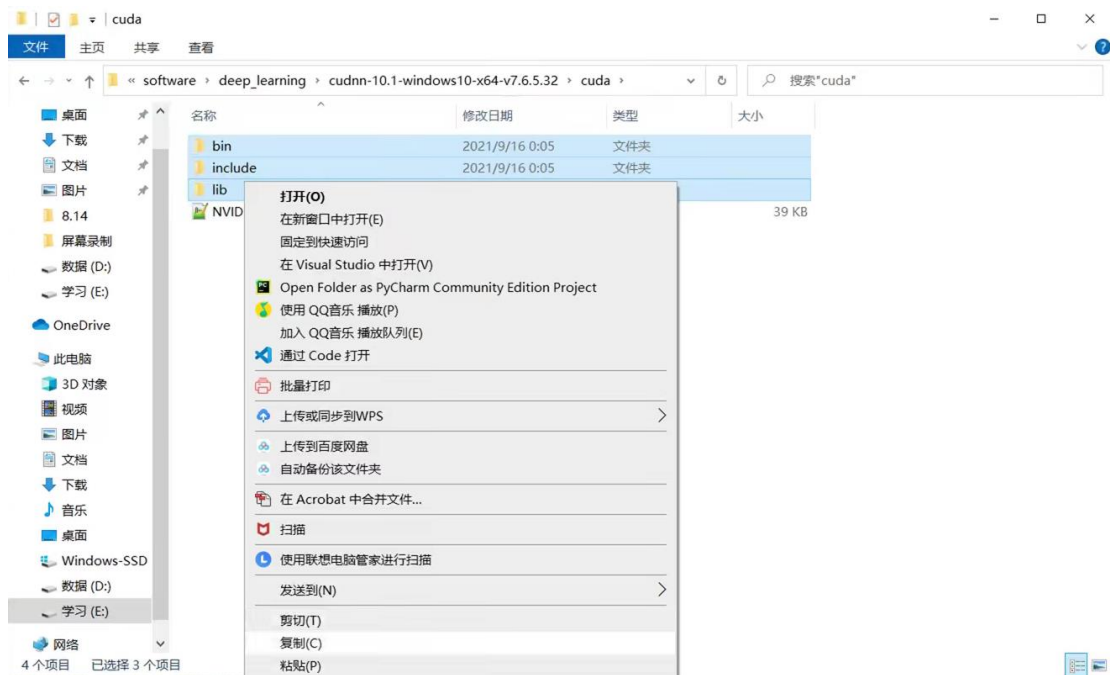
注册账号登录下载，



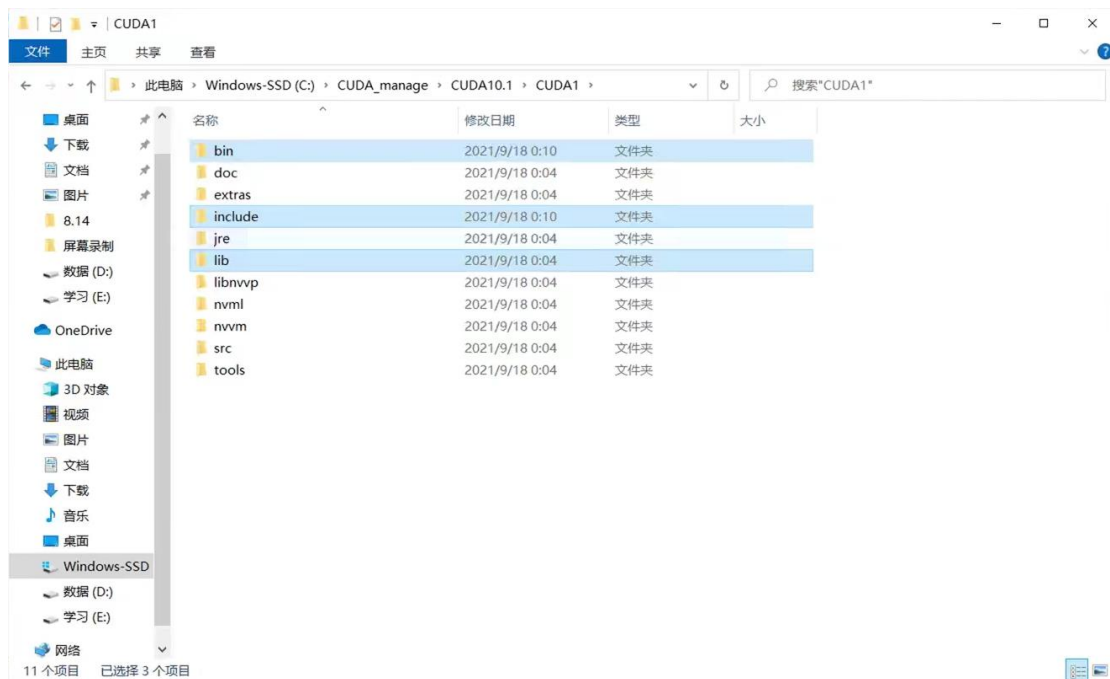
下载完成，存放在本地硬盘中，



解压文件夹，复制其中的三个文件夹 `bin`、`include`、`lib`，



将这三个文件复制到之前建立的 `CUDA1` 文件夹中，



1.6 PyTorch 的安装

PyTorch 官网: <https://pytorch.org/>

进入到需要安装 Pytorch 的环境中, 输入命令: `conda activate py37`

```
命令提示符 - conda deactivate
# $ conda deactivate

(base) C:\Users\sunqi>condaactivate py37
'condaactivate' 不是内部或外部命令, 也不是可运行的程序
或批处理文件。

(base) C:\Users\sunqi>conda activate py37

(py37) C:\Users\sunqi>python
Python 3.7.11 (default, Jul 27 2021, 09:42:29) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()

(py37) C:\Users\sunqi>conda deactivate

(base) C:\Users\sunqi>python
Python 3.8.5 (default, Sep 3 2020, 21:29:08) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()

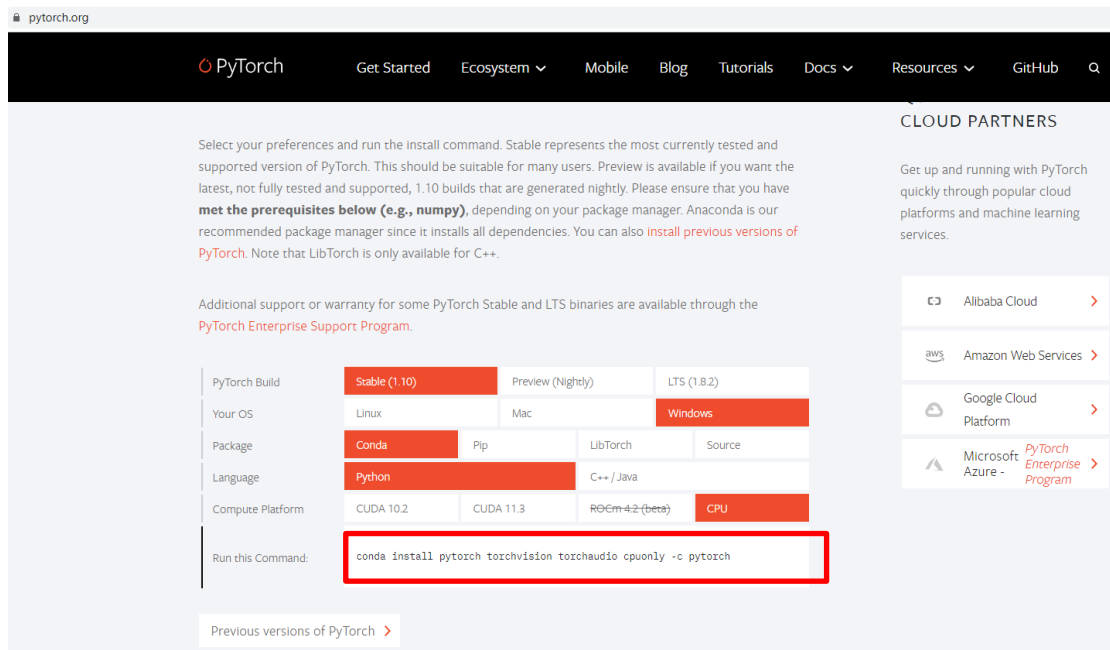
(base) C:\Users\sunqi>nvcc -V
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2019 NVIDIA Corporation
Built on Fri_Feb_8_19:08:26_Pacific_Standard_Time_2019
Cuda compilation tools, release 10.1, V10.1.105

(base) C:\Users\sunqi>conda activate py37

(py37) C:\Users\sunqi>
```

1) 安装 PyTorch CPU 版本

在 PyTorch 官网中, 选择系统对用的版本,



pytorch.org

PyTorch Get Started Ecosystem Mobile Blog Tutorials Docs Resources GitHub

Select your preferences and run the install command. Stable represents the most currently tested and supported version of PyTorch. This should be suitable for many users. Preview is available if you want the latest, not fully tested and supported, 1.10 builds that are generated nightly. Please ensure that you have met the prerequisites below (e.g., `numpy`), depending on your package manager. Anaconda is our recommended package manager since it installs all dependencies. You can also install previous versions of PyTorch. Note that LibTorch is only available for C++.

Additional support or warranty for some PyTorch Stable and LTS binaries are available through the PyTorch Enterprise Support Program.

PyTorch Build	Stable (1.10)	Preview (Nightly)	LTS (1.8.2)
Your OS	Linux	Mac	Windows
Package	Conda	Pip	LibTorch
Language	Python	C++ / Java	Source
Compute Platform	CUDA 10.2	CUDA 11.3	ROCm 4.2 (beta)
			CPU

Run this Command:

```
conda install pytorch torchvision torchaudio cpuonly -c pytorch
```

Previous versions of PyTorch >

CLOUD PARTNERS

Get up and running with PyTorch quickly through popular cloud platforms and machine learning services.

- Alibaba Cloud
- Amazon Web Services
- Google Cloud Platform
- Microsoft Azure - PyTorch Enterprise Program

在 py37 环境中输入命令: `conda install pytorch torchvision torchaudio cpuonly -c pytorch`

【此处不在演示】

2) 安装 PyTorch GPU 版本

在 PyTorch 官网中，点击“Previous versions of PyTorch”，

The screenshot shows the PyTorch website's installation guide. At the top, there's a navigation bar with links like 'Get Started', 'Ecosystem', 'Mobile', 'Blog', 'Tutorials', 'Docs', 'Resources', and 'GitHub'. Below the navigation bar, there's a section titled 'Select your preferences and run the install command.' which provides instructions on choosing between Stable, Preview, and LTS builds, and mentions prerequisites like numpy. A table below this section allows users to select their preferred build (Stable, Preview, or LTS), OS (Linux, Mac, or Windows), package manager (Conda, Pip, LibTorch, or Source), language (Python or C++/Java), and compute platform (CUDA 10.2, CUDA 11.3, ROCm 4.2 (beta), or CPU). The 'Run this Command:' section shows the command: `conda install pytorch torchvision torchaudio cudatoolkit=10.2 -c pytorch`. At the bottom, a red box highlights the link 'Previous versions of PyTorch >'. To the right, there's a 'CLOUD PARTNERS' section listing various cloud providers like Alibaba Cloud, Amazon Web Services, Google Cloud Platform, and Microsoft Azure.

选择已安装 CUDA 对应的版本，如

The screenshot shows the 'Previous versions of PyTorch' page. It provides installation instructions for different operating systems and CUDA versions. Under the 'Conda' section, there's a command: `# conda conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 -c pytorch`. Under the 'Linux and Windows' section, there are commands for different CUDA versions: `# CUDA 9.2 conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cudatoolkit=9.2 -c pytorch`, `# CUDA 10.1 conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cudatoolkit=10.1 -c pytorch`, `# CUDA 10.2 conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cudatoolkit=10.2 -c pytorch`, `# CUDA 11.0 conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cudatoolkit=11.0 -c pytorch`, and `# CPU Only conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cpuonly -c pytorch`. The command for CUDA 10.1 is highlighted with a red box. Under the 'Wheel' section, there's a command: `# CPU Only conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cpuonly -c pytorch`. Under the 'OSX' section, there's a command: `# CPU Only conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cpuonly -c pytorch`.

将命令复制到 py37 环境中：

```
conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cudatoolkit=10.1 -c pytorch
```

```
命令提示符 - conda deactivate - conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cudatoolkit=10.1 -c pytorch

(base) C:\Users\sunqi>conda activate py37
'conda activate' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

(base) C:\Users\sunqi>conda activate py37

(py37) C:\Users\sunqi>python
Python 3.7.11 (default, Jul 27 2021, 09:42:29) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()

(py37) C:\Users\sunqi>conda deactivate

(base) C:\Users\sunqi>python
Python 3.8.5 (default, Sep 3 2020, 21:29:08) [MSC v.1916 64 bit (AMD64)] :: Anaconda, Inc. on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()

(base) C:\Users\sunqi>nvcc -V
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2019 NVIDIA Corporation
Built on Fri Feb 8 19:08:26 Pacific Standard Time 2019
Cuda compilation tools, release 10.1, V10.1.105

(base) C:\Users\sunqi>conda activate py37

(py37) C:\Users\sunqi>conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cudatoolkit=10.1 -c pytorch
Collecting package metadata (current_repodata.json): \
```

安装中.....

```
命令提示符 - conda deactivate - conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cudatoolkit=10.1 -c pytorch

intel-openmp      pkgs/main/win-64::intel-openmp-2021.3.0-haa95532_3372
jpeg              pkgs/main/win-64::jpeg-9b-hb83a4c4_2
libpng            pkgs/main/win-64::libpng-1.6.37-h2a8f88b_0
libtiff           pkgs/main/win-64::libtiff-4.2.0-hd0e1b90_0
libuv             pkgs/main/win-64::libuv-1.40.0-he774522_0
lz4-c             pkgs/main/win-64::lz4-c-1.9.3-h2bbff1b_1
mkl               pkgs/main/win-64::mkl-2021.3.0-haa95532_524
mkl-service       pkgs/main/win-64::mkl-service-2.4.0-py37h2bbff1b_0
mkl_fft           pkgs/main/win-64::mkl_fft-1.3.0-py37h277e83a_2
mkl_random        pkgs/main/win-64::mkl_random-1.2.2-py37hf1a4ad_0
ninja             pkgs/main/win-64::ninja-1.10.2-h6d14046_1
numpy             pkgs/main/win-64::numpy-1.20.3-py37ha4e8547_0
numpy-base        pkgs/main/win-64::numpy-base-1.20.3-py37hc2deb75_0
olefile           pkgs/main/win-64::olefile-0.46-py37_0
pillow            pkgs/main/win-64::pillow-8.3.1-py37h4fa10fc_0
pytorch           pkgs/main/win-64::pytorch-1.7.1-py3.7_cuda101_cudnn7_0
six               pkgs/main/noarch::six-1.16.0-pyhd3eb1b0_0
tk                pkgs/main/win-64::tk-8.6.10-he774522_0
torchaudio        pkgs/main/win-64::torchaudio-0.7.2-py37
torchvision       pkgs/main/win-64::torchvision-0.8.2-py37_cu101
typing_extensions pkgs/main/noarch::typing_extensions-3.10.0.0-pyhca03da5_0
xz                pkgs/main/win-64::xz-5.2.5-h62ded97_0
zlib              pkgs/main/win-64::zlib-1.2.11-h62ded97_4
zstd              pkgs/main/win-64::zstd-1.4.9-h19a0ad4_0

Proceed ([y]/n)?
Preparing transaction: done
Verifying transaction: \
```

安装完成。

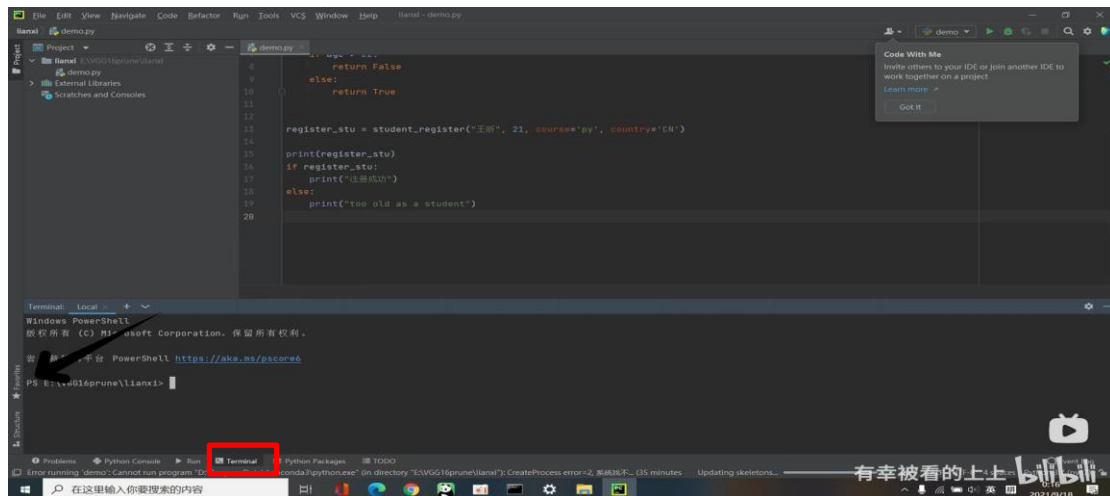
```
命令提示符 - conda deactivate - conda install pytorch==1.7.1 torchvision==0.8.2 torchaudio==0.7.2 cudatoolkit=10.1 -c pytorch

done

(py37) C:\Users\sunqi>
```

3) 测试 PyTorch 安装是否成功

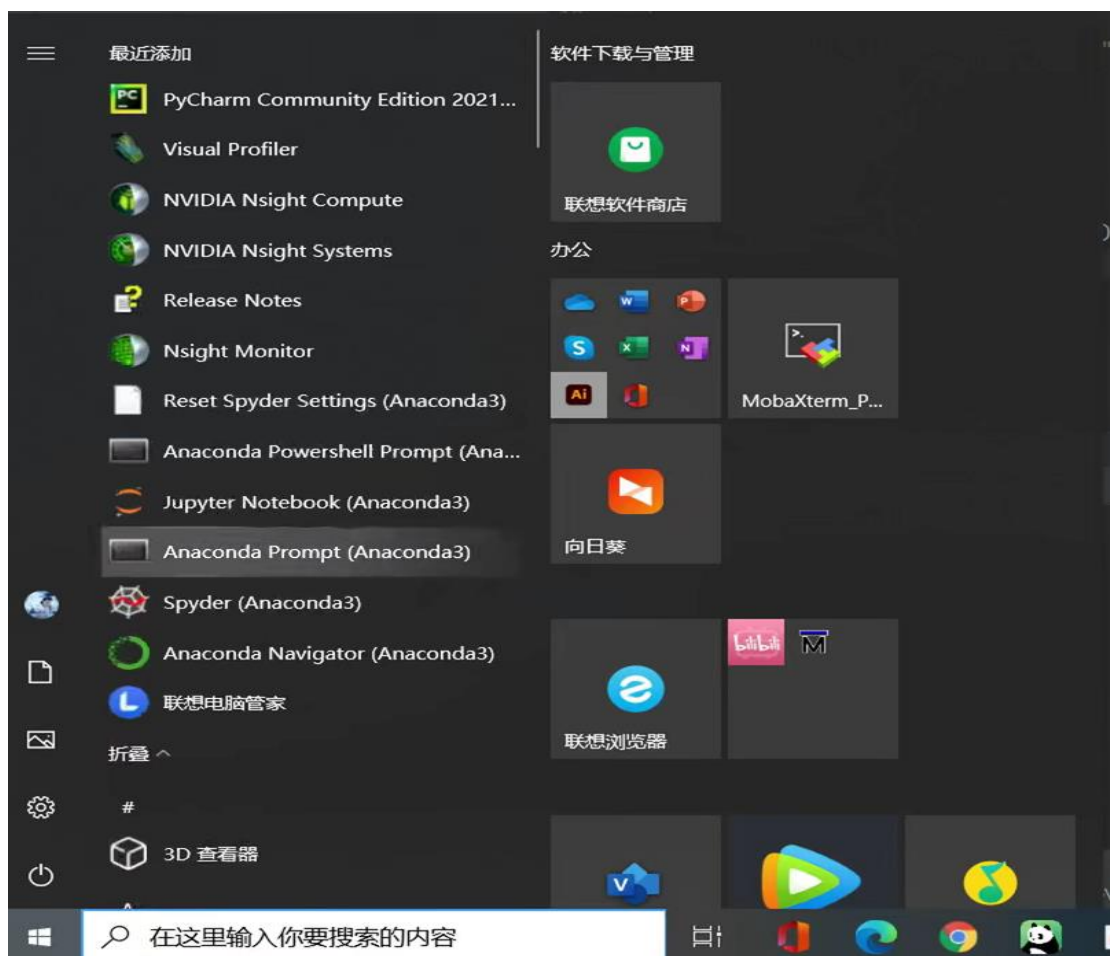
打开 PyCharm 的 Terminal,



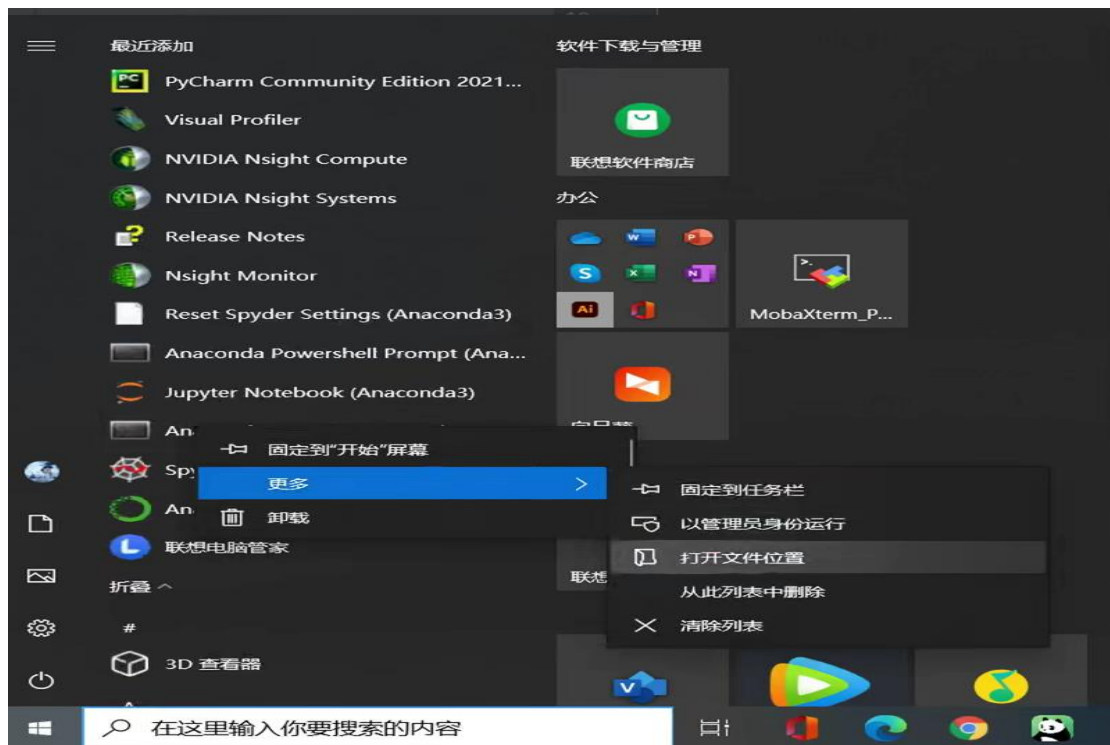
【补充】:

此处的 PS 显示有问题, 需要手动修改。

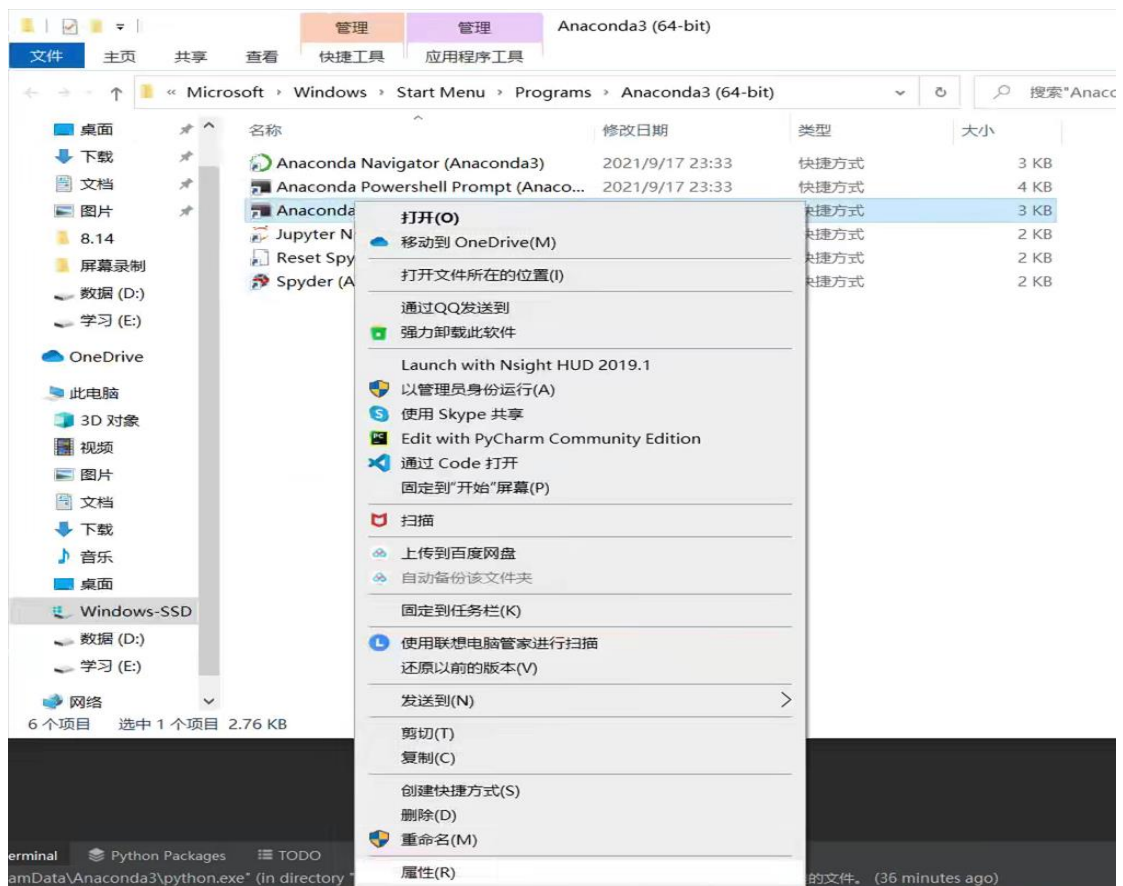
在电脑“开始”栏中, 找到“Anaconda Prompt (Anaconda3)”。



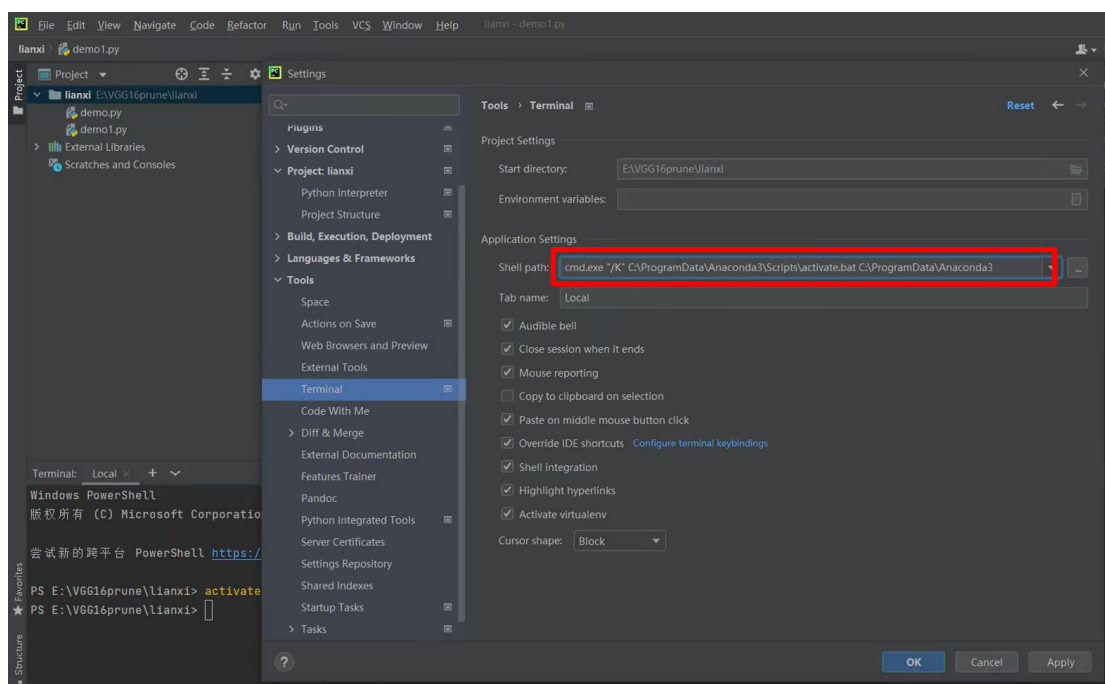
选择“打开文件位置”，



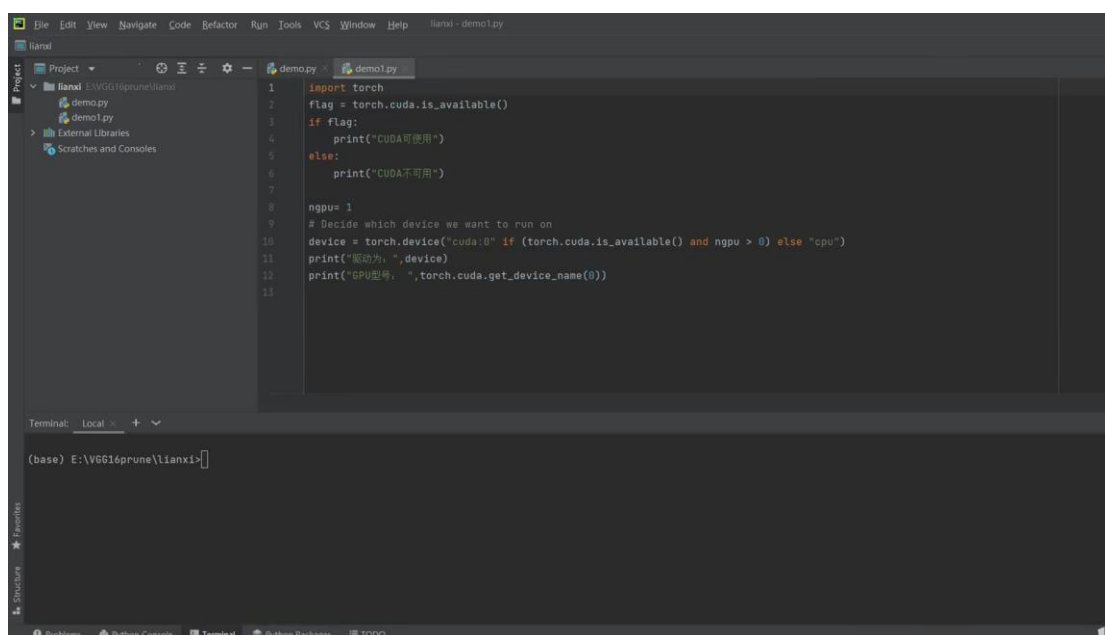
右击查看“属性”，



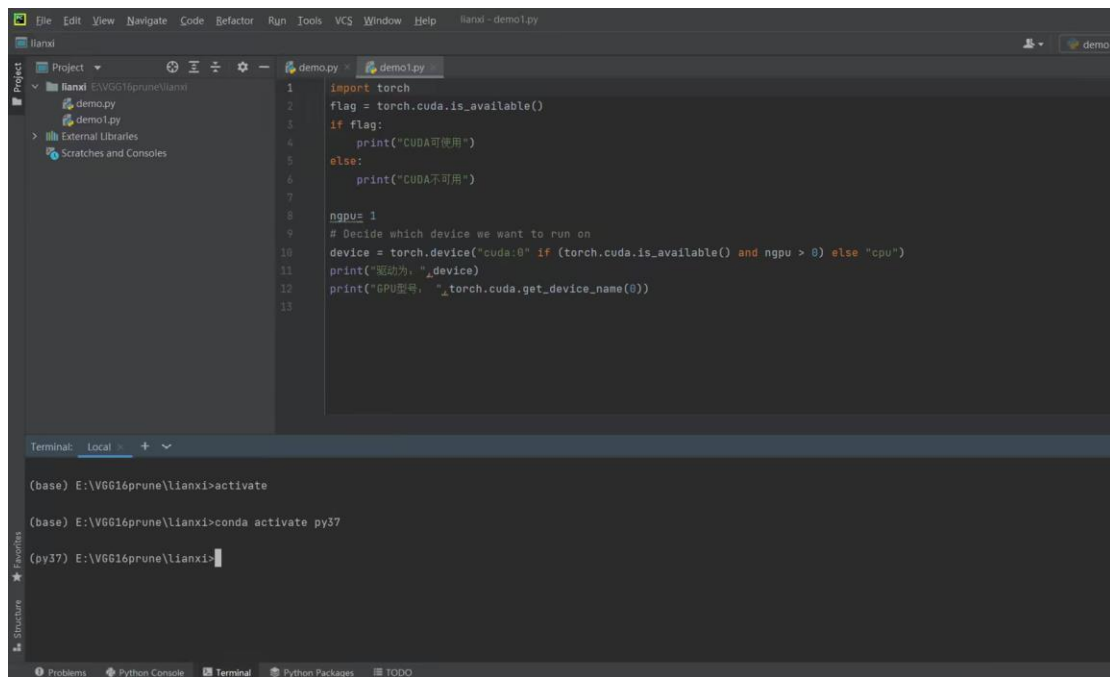
复制完地址后，点击“OK”，



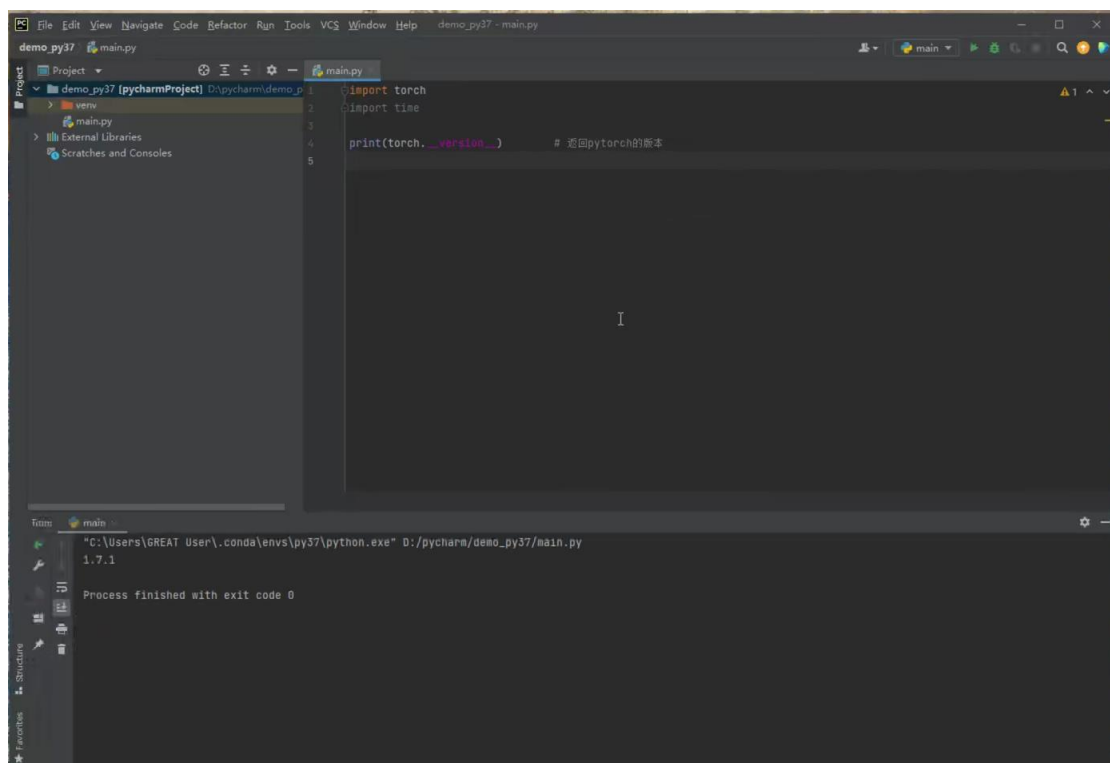
重新打开 PyCharm，点击 Terminal，则可以进入 base 环境，



激活 py37 环境，



在 PyCharm 中新建文件，输入代码，验证 torch 是否安装成功。



代码中输入 `print(torch.cuda.is_available())`

若输出为 `True`，程序就可以在 GPU 上运行

1.7 TensorFlow 的安装

TensorFlow 官网地址: <https://tensorflow.google.cn/>

安装教程待更新……

二、Windows 系统下深度学习环境配置答疑

参考视频: <https://www.bilibili.com/video/BV1Lv41177BW?p=2>

文档整理待更新……

三、Ubuntu 系统下深度学习环境配置

参考视频:

[https://www.bilibili.com/video/BV16a4y157ba?from=search&seid=10345502798454121390
&spm_id from=333.337.0.0](https://www.bilibili.com/video/BV16a4y157ba?from=search&seid=10345502798454121390&spm_id_from=333.337.0.0)

文档整理待更新……